FRC 2026 REBUILT -- Top-30 Robot Design Deep Dive

A study reference for effective design & implementation, drawn from Chief Delphi, David's Design Server (frcdesign.org), team CAD/code releases, Behind the Bumpers, and Statbotics EPA.

The 2026 design landscape

The game

REBUILT is a fuel (6" foam ball) shooter: collect fuel and score it into your hub when active. Robots may hold unlimited fuel. The end-game tower climb existed but was LOW VALUE -- most top teams dropped or never built a climber, so 2026 design is about the fuel cycle: intake → storage/indexing → launch.

The defining meta

The season split into two offensive families: (1) high-volume 'DUMPERS' -- huge hoppers that drive to the hub and empty fuel in bulk during scoring windows (1678 'Limestone', 254, 7769); and (2) 'SHOOTERS' -- flywheel or drum launchers, frequently TURRETED with vision auto-aim for shoot-on-the-move (4414, 6329, 1987, 2767, 5000, 4481). Several teams REBUILT mid-season from a passer/multi-wide shooter to a dumper (2910 Blitz→ReBlitz; 9470 triple→drum; 5687 'Big Dumpy').

Indexing was where teams innovated

Moving fuel by geometry beat squeezing it. Spindexers ('car wash' dual-stack on 1796, zero-compression drum on 6329, cone-in-bowl on 9483), drum/revolver magazines, and roller-floor serializers (1678, 1768) dominated. Large flow-through hoppers (50–80 fuel) were common.

Drivetrain & vision

Swerve is universal at this tier. Vision was a differentiator but was rarely published: confirmed stacks include 9470 (2× OV2311 global-shutter + Orange Pi 5 Pro, shoot-on-the-move derived from 6328), 1678 (Limelight 4 + CANrange), 3467 (custom Jetson + 4 cameras), 5000 (PhotonVision), 157 (PhotonVision on Orange Pi 5). Most teams left vision hardware unstated -- marked 'unconfirmed' here, meaning not published, not absent.

How to use this document

Profiles are ordered by Statbotics EPA (season-cumulative scoring contribution). For each team we capture archetype, the four subsystems, vision, public CAD/design availability, and 'lessons for students'. The teams with the most reproducible documentation for a hands-on study are 1678, 4414, 1690, 7769, 6329, 6800, 3467, 9470, 5000, and 4481. ACCURACY NOTE: subsystem specifics (motors, gear ratios, materials) are compiled from each team's public CAD, tech binder, reveal, and code, and may contain transcription or interpretation errors -- always confirm a detail against the team's own linked source before relying on it. Corrections from teams are applied verbatim (e.g., Team 5000's entry was corrected on 2026-07-06 at the team's request).

Top-30 at a glance (by Statbotics EPA)

#	Team	Robot	EPA	Rec	Archetype	Public CAD/Design
1	*** 4414 HighTide**	RIPCURRENT	**357**	70-2	Single-turret flywheel shooter + 'dye-rotor' centering; high-B	Tech binder
2	**254 The Cheesy Poofs**	Overload	**328**	65-6	'Overload': 4-wide DUMPER with an articulating hood. Chose dum	Team site
3	*** 7769 The CREW**	CHUNK	**312**	70-5	Wide single-stream 'dumper' (chose dumper over turret) with a	CD thread
4	**1323 MadTown Robotics**	(undisclosed)	**310**	69-4	Dye-rotor (drum) turret shooter + four-bar linkage intake; 202	Stats/TBA
5	**2056 OP Robotics**	(2026 binder pending)	**302**	63-10	Single-ball turret shooter fed by a high-throughput conveyor +	Video
6	*** 1690 Orbit**	KEPLER	**295**	47-5	'Kepler' trench-bot built on 4 pillars: shoot-on-the-move, lar	CAD release

#	Team	Robot	EPA	Rec	Archetype	Public CAD/Design
7	**27 Team RUSH**	(undisclosed)	**295**	83-6	Wide 3-fuel STEEL-DRUM shooter fed by a roller floor + passive	Stats/TBA
8	*** 1114 Simbotics**	Simbot Tim	**288**	62-7	Wide DRUM dumper (deliberately NO turret); huge fuel capacity.	Team site
9	**2481 Roboteers**	(undisclosed)	**285**	44-8	'Big dumper': fixed 4-wide shooter + 5-ball-wide touch-it-own-	Stats/TBA
10	**5687 The Outliers**	Glenn / 'Big Dumpy'	**283**	62-12	Fast drum-shooter that iterated into a dumper ('Big Dumpy')	Code (GitHub)
11	*** 2910 Jack in the Bot**	Blitz → ReBlitz	**282**	53-23	Built a passer/3-wide shooter ('Blitz'), then REBUILT mid-seas	CAD release
12	**125 NUTRONs**	FLUX	**282**	68-17	'Dye rotor' ('dish') shooter - NUTRONs originated the dye-roto	Team site
13	*** 9470 Ctrl-Alt-Defeat**	(triple → drum)	**279**	60-13	Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+A	Tech binder
14	**7558 ALT-F4**	(undisclosed)	**279**	51-17	Drum shooter with an advanced vision/simulation software progr	Team site
15	*** 1678 Citrus Circuits**	Limestone	**278**	61-9	World-record DUMPER / hooded shooter; led the 'Certified Dumpe	CAD release
16	**6369 Mercenary Robotics**	(full-ramp intake bot)	**274**	77-9	Rebuild from a TURRET to a wide aluminum DRUM shooter; slapdow	Code (GitHub)
17	*** 6800 Valor**	Downpour	**268**	62-10	Elevator-launched fuel scorer with assisted-aim and bidirectio	Onshape CAD
18	*** 3467 Windham Windup**	(Open Alliance)	**266**	75-10	High-throughput dumper with auto-aim shooter backup; software/	Onshape CAD
19	**9128 ITKAN Robotics**	Triple Threat	**265**	60-7	Trench robot: cycloidal slapdown intake -> duct-tape conveyor	Video
20	**1796 RoboTigers**	Sprout	**263**	48-7	'Car Wash' spindexer fuel scorer	CD thread
21	**9483 Overcharge**	Enigma	**261**	38-6	Minimalist, low-motor-count spindexer scorer	Video
22	*** 6329 The Bucks' Wrath**	Roman I / Roman II	**260**	61-15	REBUILD story: 'Roman I' (turret + low-compression spindexer,	Onshape CAD
23	*** 2046 Bear Metal**	Bearricade MkII	**260**	78-16	'Big dumper' with a hopper-floor + drum shooter; simplicity-ma	Tech binder (Drive)
24	*** 1987 Broncobots**	Cyclone	**259**	54-6	'Cyclone': 360deg turret dye-rotor shooter fed by a linear dep	CAD release
25	**1768 Nashoba Robotics**	Nightshift	**259**	70-17	Fixed full-width high-throughput floor-loader; volume shooter,	CD thread
26	**8044 Denham Venom**	Cat-5	**257**	50-9	Ran two robots: v1 (low 'bump'/under-trench) and v2 (a HighTid	DDS thread
27	*** 1706 Ratchet Rockers**	Mirage	**256**	45-2	Dual-turret shooter ('Mirage' - 'shooting so fast you think yo	CD thread
28	**9496 LYNK**	Matterhorn	**254**	70-13	2-wide drum 'over-bumper reverse' shooter (modeled on 254's 20	CD thread
29	**5460 Strike Zone**	Helios	**252**	57-11	Drum shooter mounted on a turret (a 'turret dumper'), ~30deg t	CD thread
30	*** 581 Blazing Bulldogs**	Rubble	**251**	47-19	'Rubble': dye-rotor turret shooter w/ 35-deg adjustable hood,	Onshape CAD

* = recommended study pick (high performance + a usable public CAD or design release).

Team-by-team profiles

#1 (study pick) | **4414 HighTide** RIPCURRENT | **EPA 357**
San Diego, CA | 70-2

Single-turret flywheel shooter + 'dye-rotor' centering; high-BPS single stream

SHOOTER 4× Kraken X44 on a 3" flywheel; turret geared 55.7:1 with a flex-wheel cable-chain tensioner; low gearing keeps launch current low	**INTAKE** Over-bumper/ground intake, 2× Kraken X60 + 1× X44 bottom bar; left/right sides decoupled for durability
INDEXER / HOPPER 'Dye rotor' optimized for high-BPS single-stream feeding; structural bumpers double as hopper walls AND the shooter mount	**DRIVETRAIN** 25×32" swerve, 2 custom modules from WCP Swerve X2 parts, 7.67:1; rear corners chamfered to grow hopper volume
ENDGAME Climber not detailed in the binder.	**AUTONOMOUS** High-BPS single-stream turret with vision auto-aim / shoot-on-the-move (controls-led; precomputed trajectory maps).
VISION Limelight always pointed at the hub (reads the hub's 4 AprilTags), accounting for ~95% of on-robot localization [BtB transcript]	**PUBLIC CAD / DESIGN** Interactive tech binder (no downloadable CAD)

LESSONS FOR STUDENTS

- Make structure multi-functional: bumpers serve as hopper walls and shooter mount, saving weight and volume.
- Low flywheel gearing minimizes current spikes at launch -- protects the whole electrical budget.
- Decoupling the two intake sides improves survivability against contact/defense.
- An AI-assisted state-machine codebase + interactive binder is a documentation model.

Tech binder: https://2026.team4414.com/ · Tech binder · Reveal · Statbotics · Behind the Bumpers (full walkthrough)

#2 254 The Cheesy Poofs Overload

San Jose, CA

EPA 328

65-6

'Overload': 4-wide DUMPER with an articulating hood. Chose dumper over turret at their 12-hr kickoff build (turret ceiling ~18 BPS vs ~30-35 for a 4-wide multi-stream); robustness-first. ~25-28 BPS on fresh balls. Curie Division winner [BtB transcript]

SHOOTER 4-wide dumper shooter with an articulating hood and a 'triangle of compression' - each ball contacts the main drum, the bottom feeder roller, and the top hood roller at once -> zero bobble, full spin-up, consistent shots. Only ~1.25in clearance per side, so a unique DOUBLE-SIDED belt drives the main drum (outer) and the back hood rollers (inner) for the reversing effect; backing feeder swapped poly -> CatTongue (poly slipped); shooter motors nested behind [BtB transcript]	**INTAKE** Most-iterated subsystem. Rack-and-pinion deploy: racks went from 3/8in polycarb w/ titanium teeth (kept breaking) to TWO layers of 1/4in SRP + one 1/8in titanium (unbreakable). A bottom 3-roller rips balls up TITANIUM kicker plates/bars (aluminum bent) tuned for low compression without dead spots; a 2nd roller keeps stored hopper balls from blocking incoming ones; dual Kraken X60. (Used GoPros inside to find jams; SRPP chosen to spread load around bolt holes, ~30% lighter, impact-resistant) [BtB transcript]
INDEXER / HOPPER 'Bumper hopper' = a full 8in polycarb wall extending up from the bumpers (more width, fewer parts) + a carbon-fiber hoop/corners for rigidity; a NET raises capacity from ~50 to ~75-80 balls. Hopper floor = WCP rubber-back timing belts (smoother than rollers). The floor's pivot is LINKED to the intake deploy ('trash compacting'): pulling the intake in while shooting lifts the floor so gravity empties the last balls; the intake fully stows under the floor [BtB transcript]	**DRIVETRAIN** Swerve (module not detailed); deliberately does NOT shoot on the move (current limits) - instead optimizes for fast alignment + a defined pivot point [BtB transcript]
ENDGAME Climber not detailed (dumper archetype).	**AUTONOMOUS** Dumper: drive to the hub and empty fuel in bulk during scoring windows; auto routines not detailed.
VISION FOUR cameras (their most ever) on ONE Jetson Orin Nano with GPU-accelerated AprilTag detection: two grayscale 120fps on the shooter side (heavy lifting) + two 60fps side cameras; MULTI-TAG only (rejects single-tag), ~100+ fps. The Jetson solves and sends poses over UDP to the RIO, which filters by reprojection error + ambiguity. Heading control uses a predicted look-ahead pose + feed-forwards + an intelligent center-of-rotation (pivots about a corner for fastest alignment); Choreo 3-stage autos with a recovery path [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD (reveal video + build blog only)

LESSONS FOR STUDENTS

- 254's build blog shows disciplined prototyping (backspin single-wheel vs dual flywheel) before committing.
- Even the #2 team in the world withheld CAD -- when a team publishes nothing, match video is your scouting source.
- Their strategic game-analysis method is as worth studying as the hardware.

Team site: https://www.team254.com/first/2026/ · 254 2026 · Reveal · DDS robot-pics-2026 · Behind the Bumpers (full transcript)

#3 (study pick)	**7769 The CREW** CHUNK Michigan	**EPA 312** 70-5

Wide single-stream 'dumper' (chose dumper over turret) with a high-capacity hopper

SHOOTER Evolved from a two-chamber shooter to a 2.75"-wide shooter; praised for distance + accuracy. The hood surface is static rather than rollered, which removes the external belt that would otherwise run outside the hood plates and be damaged when the robot is pushed into the trench; with the hood up the plates react against the trench instead of letting the robot be pushed underneath. Fuel does not jam at the 4 in stealth wheels because their surface speed is far higher than the feeding wheels. After a partial rebuild between the week 2 and week 4 events, powering both the top and bottom indexing rollers pushed fuel into the chamber too fast and jammed it; the fix was to fix the lower rollers so they cannot rotate, run the top rollers powered against a static lower surface, and that configuration ran the rest of the season. Per the team in the release thread.	**INTAKE** Linear slapdown intake; deploy arms cut from polycarbonate rack + 98A TPU printed chain tensioner; dynamic current limit (45A deploy → 5A hold) for compliance. Each side is driven independently. Per side there are four 1/8 in polycarbonate plates, two with a flat bottom to slide on and two cut for the WCP 10dp racks, giving 1/2 in of slide width; the same 10dp aluminium gears ran all season and the racks were replaced once. The rack is fixed to the intake with heat-set inserts rather than threading into the polycarbonate, which survived repeated full-speed rams. The TPU idler is not strictly required: exact centre-to-centre spacing ran without it for the first two events, and it was added to keep the two sides in sync. Per the team in the release thread.
INDEXER / HOPPER High capacity: ~70 fuel max, 50–60 in play; in/out throughput was their edge. The vectored wheels are print-in-place, adapted from a 2910 model with the roller-to-shaft clearance opened up so the rollers spin reliably and the wheel slips onto hex, plus a 1/8 in boss on one side so wheels stack without a spacer; printed in PETG, one set lasted the whole season because they are protected inside the frame. The belt floor uses stock 105T x 15 mm timing belts. Per the team in the release thread.	**DRIVETRAIN** Swerve; module make and ratios are not stated in the release thread. Read them from the Onshape CAD.
ENDGAME The robot has a climber: the robot name came from a part on it that was mis-named in CAD. Mechanism specifics are not given; unconfirmed.	**AUTONOMOUS** Drive-and-empty dumper cycle. Auto routines are not detailed in the release thread.
VISION Not stated in the release thread or the tech slides; unconfirmed. Check the CAD for camera mounts.	**PUBLIC CAD / DESIGN** Public CAD + tech slides (Onshape)

LESSONS FOR STUDENTS

- Massive hopper capacity + high throughput beat turret complexity for them -- a clear dumper case study.
- Candid failure analysis: polycarb intake teeth sheared every 2–3 matches at champs; they ran ~16 spares. Pick intake materials for fatigue.
- Software current-limit tuning can buy you mechanical compliance without extra hardware.
- A stated design rule, applied every year: build the simplest robot that meets the objectives the team decided are needed to be competitive. The static hood is that rule applied to one part.
- Their jam fix went the opposite way to intuition: they removed drive from the lower indexing rollers rather than adding more power, because the jam was fuel arriving too fast, not too slow.

CD thread: https://www.chiefdelphi.com/t/frc-7769-cad-tech-slides-chunk/521008 · CAD + slides · Behind the Bumpers

#4 1323 MadTown Robotics (undisclosed)

Madera, CA

EPA 310

69-4

Dye-rotor (drum) turret shooter + four-bar linkage intake; 2026 World Champion (20-1 at Champs) [FRC StratAnalyst video - secondary source]

SHOOTER Dye-rotor (drum) turret (motors/gearing unconfirmed) [FRC StratAnalyst video]	**INTAKE** Four-bar linkage ground intake (details unconfirmed) [FRC StratAnalyst video]
INDEXER / HOPPER Unconfirmed	**DRIVETRAIN** Swerve (assumed)
ENDGAME Unconfirmed (minimal public documentation).	**AUTONOMOUS** Unconfirmed (minimal public documentation).
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD/design release

LESSONS FOR STUDENTS

- Strong results (#4, 69-4) don't require public documentation -- design knowledge isn't always shareable.
- When a team publishes nothing, use TBA match video as the primary scouting source.
- A candidate to contact directly rather than study from public CAD.

Stats/TBA: https://www.thebluealliance.com/team/1323/2026 · Team site · TBA 2026 · FRC StratAnalyst top-10 video

#5 2056 OP Robotics (2026 binder pending)

Stoney Creek, ON

EPA 302

63-10

Single-ball turret shooter fed by a high-throughput conveyor + ~62-ball net hopper; world-record scorer

SHOOTER Turret-mounted single-ball shooter: 2x Kraken X60 flywheel run at ONE fixed RPM (never varied up/down), hood on one Kraken X44; 370deg rotation via a sprung, non-bidirectional cable chain [BtB transcript]	**INTAKE** Linear EXTENDING intake with Lexan flex side-plates (survives collisions, enters opponents' zone); front rollers on a Kraken X60 with plain-rubber coating; surgical-tubing kicker bar added mid-season to stop balls sticking against the bumper; 3D-printed wire-management chain [BtB transcript]
INDEXER / HOPPER Neoprene-coated 'hot dog' rollers + a live-bottom funnel; a bare-Lexan 'slip' roller before the conveyor as an anti-jam; conveyor = top+bottom timing belts (thicker WCP bottom belt) compressing balls up into the turret; a reverse-spinning anti-jam roller; net hopper (~62 fuel) repurposed from prior years [BtB transcript]	**DRIVETRAIN** Trapezoidal swerve base (per Behind the Bumpers)
ENDGAME Climber not detailed in the public material.	**AUTONOMOUS** Single-ball turret + high-throughput conveyor; auto/vision detail in their Behind the Bumpers video.
VISION Not named in the Behind the Bumpers walkthrough (throughput is conveyor-driven with a single-speed flywheel); camera/coprocessor unconfirmed	**PUBLIC CAD / DESIGN** 2025 binder public; 2026 not yet released

LESSONS FOR STUDENTS

- 2056's 'Behind the Bumpers' series is a model for student-facing design explanation -- watch the 2026 entry.
- Their prior technical binders (e.g. 2025) are a template for a student design study.
- Research trap to avoid: 'LIGHTNING' is their 2025 robot, not 2026 -- verify season before citing.

Video: https://www.youtube.com/@2056public · Team site · YouTube · TBA 2026 · Behind the Bumpers (read full transcript)

#6 (study pick)	**1690 Orbit** KEPLER Israel	**EPA 295** 47-5

'Kepler' trench-bot built on 4 pillars: shoot-on-the-move, large (~60-ball) hopper, extreme durability, low profile; roller-floor indexer into a 640deg turret. Galileo Division winner [BtB transcript]

SHOOTER
Compact 640deg turret (360+ per side): X60 drive through a low gearbox (low CG), self-made gears on an 8in bearing, ball path offset toward the back to tuck into the corner. Shooter tuned for slight/no spin (most consistent stream, lowest energy loss): back rollers belted to a custom main shooter wheel; motors on an SLS winch pulled tight to the wheel + hood motors on the plate side via bevel gears to MINIMIZE diameter; custom 3D-printed energy chain + 14ga wires + a raised 'neck' so the chain wraps tight enough for 640deg [BtB transcript]

INTAKE
Durable/collapsible/swappable wide intake: two silicone-coated rollers on a Kraken X60 + a static bottom kicker roller; deliberately NO slapdown (poor impact collapse) - instead an in-house RACK-AND-PINION deploy (polycarb racks + aluminum/brass gears); 25x50mm front extrusion + a polycarb/aluminum side sandwich; the whole U-frame lifts off via 3 screws per side for fast swaps [BtB transcript]

INDEXER / HOPPER
3 stages: (1) roller floor of 11 carbon-fiber silicone rollers on an X60 (2:1), roller-to-roller via 3mm GT2 belts, a slight reduction to ease the 4-wide -> 3.5-wide jam point, plus two un-coated rollers to cut friction; (2) two 30mm 5mm-HTD vertical belts on two independent X44 (~8 m/s) over a 1mm PTFE sheet (~10mm compression) + a vertical kicker roller (repurposed from their 2024 Doppler shooter) on an X44 to clear jams; (3) an active 'C' shooter ramp (3 rows of 2.25in flex wheels + 2 rows of 2in stealth wheels) on an X60 -> 8 m/s with ZERO spin into the turret [BtB transcript]

DRIVETRAIN
29.5x24.5in (wide side = intake); bumpers attach to the hopper walls (thin polycarb + a net for ~60-ball capacity). MID-SEASON swap from a custom MK5-based module to stock SDS MK5 modules after the defense bot pushed them around - the MK5's wide spike treads dig into the carpet vs their thin nitro treads; reinforced/integrated top plates. 'Defense mode' X-locks the swerve [BtB transcript]

ENDGAME
Low-profile 'trench-runner' geometry trades climb access for faster, safer collection -- climbing is not a focus.

AUTONOMOUS
Shoot-on-the-move is one of the four stated design pillars (see CAD/reveal).

VISION
A camera ON THE TURRET on an Orange Pi running PhotonVision (aimed at the hub's AprilTags) + two rear cameras for blind spots when the turret cable wraps or a camera drops; tilt + bump detection lets it shoot while tilted; an off-robot shoot-on-the-move simulation (gravity, air resistance, Magnus) precomputes trajectory polynomials the robot evaluates live [BtB transcript]

PUBLIC CAD / DESIGN
Full public CAD (Onshape + x_t)

LESSONS FOR STUDENTS

- Choosing a low-profile/trench geometry trades climb access for faster, safer ball collection.
- First full year on Onshape -- a good case study in CAD-tool migration and clean release practice.
- A full-floor roller intake maximizes pickup area and tolerates sloppy approaches.

CAD release: https://www.chiefdelphi.com/t/frc-orbit-1690-2026-robot-cad-release/520673 · CAD release · Reveal 'Kepler' · Video · Behind the Bumpers (full transcript)

#7 27 Team RUSH (undisclosed)

Clarkston, MI

EPA 295

83-6

Wide 3-fuel STEEL-DRUM shooter fed by a roller floor + passive hopper; high-throughput dumper. Hall of Fame; Michigan State champion; ~330 EPA, 113 lb [BtB transcript]

SHOOTER One main STEEL 3in drum run by 4x Kraken X60 + two 1in hood rollers on the same X60s (spin ~2x faster to minimize backspin); 1x X44 rack-and-pinion hood; mostly plastic SRP + carbon-fiber structure for weight. VARIABLE compression (aluminum rods ~5/8in, CF stability rods ~3/8in, final ~1/2in) to fit a retrofit shooter compactly; the feeder accelerates fuel BEFORE the shooter, so the shooter only adds the final boost [BtB transcript]	**INTAKE** Main roller on 2x Kraken X44 (2.5:1) + a free-spinning carbon-fiber top roller + 3 passive inner tubes; built for quick fixes (it's at the robot's fighting face); the HOPPER is driven PASSIVELY off the intake [BtB transcript]
INDEXER / HOPPER Roller floor + indexer on 2x Kraken X60; floor = 1in polycarb rollers (5 with CatTongue tape) funneling to the 3-wide shooter via dead-axle 3D-printed guides; geared 1:1 in front, ~17:19 at the back to speed balls up; a free-spinning roller prevents jams (replaced a belted one). Iterated from belt/'sushi'/chain to all-polycarb rollers [BtB transcript]	**DRIVETRAIN** Swerve; ALL electronics mounted on the bottom (frees space for the roller floor + shooter); the electrical cover attaches to the BUMPERS so its weight doesn't count against inspection; 3D-printed component mounts [BtB transcript]
ENDGAME Unconfirmed -- reveal is announcement-only.	**AUTONOMOUS** Unconfirmed -- reveal is announcement-only.
VISION Main Limelight 4 (heavily weighted, accepts tags mainly from the rear given the shooting orientation) + left & right Limelights fused as one entity into the pose estimate for odometry during feeding / when the back can't see [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD/design release

LESSONS FOR STUDENTS

- An 83-6 record (highest win count in the top-30) with no public release: execution and consistency can outweigh documentation.
- Reveal video + match footage are the only primary design sources.
- Michigan's competitive density (27, 7769, 5460...) is itself a design driver -- strong regions force strong robots.

Stats/TBA: https://www.thebluealliance.com/team/27/2026 · Reveal thread · Video · TBA · Behind the Bumpers (full transcript)

#8 (study pick) 1114 Simbotics Simbot Tim

St. Catharines, ON

EPA 288

62-7

Wide DRUM dumper (deliberately NO turret); huge fuel capacity. Hall of Fame; won both Ontario districts + the Ontario provincial championship [BtB transcript]

SHOOTER Big belt-driven DRUM, an X44 + 3 more motors; hood uses only ~8deg of travel. They chose a drum (more versatile) OVER a turret (didn't suit their play style) and chose not to go under the trench [BtB transcript]	**INTAKE** 2x Kraken X60; balls hit a kicker bar + main roller; an 'agitating' intake with multiple states - it can vibrate fast to push balls into the indexer harder, or run gentler [BtB transcript]
INDEXER / HOPPER Extending hopper -> a moving 'rolling floor' (two rollers) carries balls up to the drum [BtB transcript]	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME Climber not detailed (wide drum dumper, no turret).	**AUTONOMOUS** Drive-and-empty dumper; auto routines not detailed.
VISION 3x Limelight cameras with MegaTag2 + AprilTags, fused with odometry for pose estimation [BtB transcript]	**PUBLIC CAD / DESIGN** No 2026 CAD release confirmed

LESSONS FOR STUDENTS

- Simbotics is well-known for STRATEGIC DESIGN EDUCATION -- their resources/presentations are a top study reference even without 2026 CAD.
- Watch the 'Simbot Tim' Behind the Bumpers for subsystem rationale.
- Pair with 2056 (also Ontario) to compare two strong programs' 2026 approaches.

Team site: https://www.simbotics.org/resources · Showcase · Behind the Bumpers · Resources

#9 **2481 Roboteers** (undisclosed) — Tremont, IL — **EPA 285** — 44-8

'Big dumper': fixed 4-wide shooter + 5-ball-wide touch-it-own-it intake + 115-ball hopper (no turret); Milstein Division winners [BtB transcript]

SHOOTER 4-wide fixed shooter, ~20 BPS; 4 accelerator rollers for accuracy; 3x Kraken X44 on the shooter + 2x X44 + 1x X60 on feed rollers; grip-tape (skateboard tape) for consistent ball spin (+~40% accuracy), a late-season PVC-pipe guide (+~10%); 2-position linear-actuator hood (replaced a pneumatic piston to cut weight), flywheel RPM varied for distance [BtB transcript]	**INTAKE** 5-ball-wide 'touch-and-take' intake: 3in roller wrapped in grip tape on 2x Kraken X60, SRPP side plates with 'Flintstone' wheels; ball compressed between roller and ramp; anti-jam = a 2nd motor + a plate to stop the roller stalling [BtB transcript]
INDEXER / HOPPER Massive 115-ball, 5-wide hopper funneling to the 4-wide shooter through the accelerator rollers [BtB transcript]	**DRIVETRAIN** Custom swerve (designed in the 2024 offseason) with large wheels and an adjustable ratio (12-32 ft/s); geared LOW this year for torque/low current under the heavy fuel load [BtB transcript]
ENDGAME Climber not detailed (fixed 4-wide dumper, no turret).	**AUTONOMOUS** Dumper; auto routines not detailed.
VISION Not mentioned in the Behind the Bumpers walkthrough; camera/coprocessor unconfirmed	**PUBLIC CAD / DESIGN** No public 2026 release (GitHub stops at 2025)

LESSONS FOR STUDENTS

- Strong record (44-8) with a low public footprint -- another reminder strong results needn't be open-sourced.
- Study via match video; consider contacting the team directly.

Stats/TBA: https://www.thebluealliance.com/team/2481/2026 · Team site · TBA 2026 · Behind the Bumpers (full transcript)

#10 **5687 The Outliers** Glenn / 'Big Dumpy' — Portland, ME — **EPA 283** — 62-12

Fast drum-shooter that iterated into a dumper ('Big Dumpy')

SHOOTER 'Insanely quick' near-full drum shooter; high cycle rate prioritized	**INTAKE** Over-bumper roller intake -- 2" upper + 1.25" lower roller with ~½" compression (minimized to avoid bogging the motor)
INDEXER / HOPPER CTRE/Kraken 'Feeder' subsystem feeding a calculated-shot ('ShotCalculator') shooter; 3-roller intake (top/bottom rollers + deployer)	**DRIVETRAIN** Swerve (team has historically experimented with differential swerve)
ENDGAME Climber not detailed (drum -> dumper).	**AUTONOMOUS** Fast drum/dumper cycle; auto routines not detailed.
VISION Hybrid Limelight + PhotonVision: 3 cameras (front + left + right), AprilTag pose (MegaTag); C++ codebase	**PUBLIC CAD / DESIGN** Code public (GitHub, C++); no CAD release

LESSONS FOR STUDENTS

- Empirical compression tuning -- they minimized roller compression by experiment to reduce motor load.
- A separate 'LifeBot/Big Dumpy' repo signals a deliberate mid-season pivot to the dumper meta.
- A C++ codebase (less common than Java) is a good study for advanced programmers.

Code (GitHub): https://github.com/frc5687 · 'Glenn' reveal · GitHub · TBA · Robot code (C++)

#11 (study pick)	**2910 Jack in the Bot** Blitz → ReBlitz Mill Creek, WA	**EPA 282** 53-23

Built a passer/3-wide shooter ('Blitz'), then REBUILT mid-season into a dumper ('ReBlitz')

SHOOTER Blitz: 3-fuel-wide shooter, whole robot aims like a 3-wide turret, ~30 fuel/sec; ReBlitz dumper hit 30+ balls/sec. The hood is belt driven rather than chain driven because there was not room for chain at the gear ratio they wanted; the team notes a single belt or chain draws similar current, belt drag only matters where there are many of them, and chain behaves poorly at high speed. Hood plates use flathead screws because a button head would rub the hopper. Three rollers visible inside the shooter are aluminium supports that fasten the backing plates with zip ties; they began as feeder rollers and were removed after they worsened shot consistency and throughput. Between DCMP and Worlds the brass flywheel was remade with the 10T pulley integrated into it. Per the team in the release thread.	**INTAKE** Dual intakes on Blitz. Powered roller floor feeder, taken deliberately from 1678 and 6328, and chained rather than belted: the team states chain has much lower resistance, lets the whole feeder and hopper run off one chain run, and is a contributing factor in reaching 34 BPS. The intake hardstop is the bumper itself; the gap visible in CAD is there for bumper fabric thickness, and on the DCMP version they had to lower that bumper edge about 1/16 in to let the intake reach the floor. Per the team in the release thread.
INDEXER / HOPPER Blitz used a multilayered indexer + intentionally SMALL hopper (a noted tradeoff); ReBlitz went to an open dumper path. Two changes between DCMP and Worlds are credited with passing 30 BPS: the feeder and hopper were geared together on 4 motors instead of running as separate powertrains, and a Teflon-taped carbon fibre board was added in front of the hopper rollers so that fuel filling the hopper hits the board and slides down into the shooter entry. Per the team in the release thread.	**DRIVETRAIN** Swerve. Electronics sit on a brainpan behind a polycarbonate bellypan shield. On Blitz they were mounted underneath, so access meant flipping the robot and removing the shield; on Re-Blitz some moved on top of the brainpan for access. The baseplate uses a rectangular lightening pattern rather than a triangular one: the team states it saves more weight, and a belly or brain pan is not a structural member so it does not need the stiffness a triangular pattern buys. Module make and ratios are not stated; read them from the CAD.
ENDGAME Not detailed in the release thread; see the tech binders for both robots.	**AUTONOMOUS** Dumper after the Blitz -> ReBlitz rebuild; auto routines not detailed.
VISION Not detailed in the release thread. The electrical system the team describes is a roboRIO, PDP, Pigeon and a CANivore, with the swerve CANcoders, Pigeon and CANranges on one splice. There is no mini power module and no Ethernet switch; Ethernet goes straight into the radio, and the team reports no radio heat problem. Camera hardware unconfirmed; see the code release.	**PUBLIC CAD / DESIGN** Onshape CAD + tech binders for BOTH robots (CD release, 2026-06-07)

LESSONS FOR STUDENTS

- THE case study in the season's central lesson: they scrapped a working passer and rebuilt as a dumper -- and it won PNW DCMP + a return to Einstein.
- Now fully studyable: downloadable Onshape CAD AND a technical binder were released for BOTH Blitz and Re-Blitz, so you can diff the original vs the rebuild directly.
- Explicitly shows the hopper-capacity vs throughput tradeoff.
- Demonstrates the courage/value of abandoning a working robot mid-season when the meta shifts.
- The rebuild was run to a published schedule: after the first event an engineering realignment meeting concluded the original robot was too highly integrated for iteration to reach the target, then 5 days were allotted to design, 1.5 weeks to assembly and 1 week to bring-up and tuning.
- Gains between DCMP and Worlds came from all three of robot iteration, driver practice including match simulation and defense practice, and strategy work -- not from hardware alone.

CAD release: https://www.chiefdelphi.com/t/2910-cad-and-tech-binder-release-2026/521705 · CAD + Tech Binder release · Blitz reveal · Behind the Bumpers

#12	125 NUTRONs FLUX Boston, MA	EPA 282 68-17

'Dye rotor' ('dish') shooter - NUTRONs originated the dye-rotor concept (2017) and went BIGGER this year; trapezoidal drive base [BtB transcript]

SHOOTER Enlarged dye-rotor 'dish' (sized up this year for throughput) fed by a live-shaft 3in carbon-fiber drum (CatTongue tape + silicone wrap) with a 3-standoff kick bar that maintains ball compression through the path [BtB transcript]	**INTAKE** Four-bar linkage deploy, modular (whole intake pops off via 2 shoulder bolts per side); a stiffener bar for aggressive driving; the intake is carved out so the robot can sit UNDER the trench at match start [BtB transcript]
INDEXER / HOPPER Live-shaft 3in CF drum + 3-standoff kick bar feeding the dye-rotor dish [BtB transcript]	**DRIVETRAIN** Trapezoidal base - wider at the FRONT (bigger intake) and narrower at the REAR (balls rarely reach the back); 1/4in side plates (no 2x1 main structure); 2 stock WCP Swerve X2S modules at the front + 2 custom-machined X2S at the rear, shaped to the dish radius [BtB transcript]
ENDGAME Climber not detailed (dye-rotor shooter).	**AUTONOMOUS** Dye-rotor shooter; auto routines not detailed.
VISION Not mentioned in the Behind the Bumpers walkthrough; camera/coprocessor unconfirmed	**PUBLIC CAD / DESIGN** No 2026 CAD release located

LESSONS FOR STUDENTS

- A 68-17 record from a distributed 5-school co-op is a model for sustaining strong performance across a large org.
- Mechanism detail lives in the FUN 'Behind the Bumpers' episode -- watch rather than read.

Team site: https://www.nutrons.com/ · Reveal thread · Behind the Bumpers · TBA

#13 (study pick)	9470 Ctrl-Alt-Defeat (triple → drum) Half Moon Bay, CA	EPA 279 60-13

Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+ALT+DEL' deadline-driven scope control

SHOOTER Drum shooter (mid-season rebuild from a Week-1 triple shooter): open fuel path, three feeder rollers on two Kraken X60.	**INTAKE** Reinforced beater-bar intake -- a flippable 2x1 aluminum bar (flip it when bent), then two 0.25\" SRP leaf-spring plates per side with the intake tube as a hardstop and a detachable roller mount.
INDEXER / HOPPER Hopper with COTS aluminum hubs + compliant floor wheels carried over from the triple-shooter robot.	**DRIVETRAIN** Swerve; a CANivore drives 18-motor shoot-on-the-move bursts.
ENDGAME Climber not detailed -- engineering hours went to scoring and intake robustness.	**AUTONOMOUS** Empirical shot map (measured, not modeled; first-shot +250 RPM compensation from data) with shoot-on-the-move.
VISION Two OV2311 global-shutter cameras + Orange Pi 5 Pro coprocessors; SOTM / vision / current-limiting adapted from 6328 and the NINJAS; 4.6 V brownout floor so failures degrade gracefully.	**PUBLIC CAD / DESIGN** Public tech binder + GitHub

LESSONS FOR STUDENTS

- Their 'CTRL+ALT+DEL' rule ships the robot 1.5 weeks before every event -- the deadline decides what to cut, copy, or skip (applied 3x: build, rebuild, reinforce).
- Reinforce the proven failure point (the intake) instead of chasing optimization -- robustness over speed for playoffs.
- Agentic coding tools for boilerplate let juniors plan instead of writing Java from scratch.

Tech binder: https://9470techbinder.vercel.app/ · Tech binder · Engineering process · Behind the Bumpers

#14 **7558 ALT-F4** (undisclosed) — North York, ON — **EPA 279** — 51-17

Drum shooter with an advanced vision/simulation software program; rebuilt after their 2nd event; Ontario Technology Division champions [BtB transcript]

SHOOTER 3in aluminum DRUM with silicone tubing + two 1.5in silicone hood rollers; drum powered by 4x Kraken X60 geared 1:1; plus a custom laser-cut 1/2in-thick x 4in-wide STEEL flywheel [BtB transcript]	**INTAKE** Linear intake running on HTD parts with 10DP gear-tooth racks meshing 10DP gears [BtB transcript]
INDEXER / HOPPER Unconfirmed	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME Climber not detailed.	**AUTONOMOUS** Advanced vision/simulation software program drives auto-aim; routines in code.
VISION Multiple Limelights with a sophisticated localization pipeline: per-camera outlier rejection -> 'smart' MegaTag1/MegaTag2 selection via hysteresis -> staged standard-deviation calc; logs everything to simulate the robot without owning a field [BtB transcript]	**PUBLIC CAD / DESIGN** No public 2026 release located

LESSONS FOR STUDENTS

- Their 2024 turret-on-swerve + efficient diverter suggests strong mechanism-aiming experience.
- 2026 design is video-only -- treat as a watch-the-reveal study.

Team site: https://www.team7558.com/ · Team site · YouTube · TBA · Behind the Bumpers (full transcript)

#15 (study pick) **1678 Citrus Circuits** Limestone — Davis, CA — **EPA 278** — 61-9

World-record DUMPER / hooded shooter; led the 'Certified Dumper Group' with 254

SHOOTER Hooded flywheel fed from a full-width 3.5" aluminum drum (grip-taped); hood on an X44 jackshaft for a wide shot-angle range	**INTAKE** Full-width slapdown ground intake -- a ~6 lb aggressive over-bumper roller (Kraken X60 pivot, silicone roller + kicker bar)
INDEXER / HOPPER Roller floor (dead-axle + flex-wheel rollers) → ball tunnel/serializer; expanding hopper; pragmatic 'strategically placed wood' guides	**DRIVETRAIN** Swerve, SDS MK5n, 27×27"
ENDGAME Climber not detailed (world-record dumper / hooded shooter).	**AUTONOMOUS** Drive-and-empty dumper; auto routines not detailed.
VISION Limelight 4 (single rear unit) + 2× CANrange sensors; custom localization + a driver dashboard with a 'shift' system (CONFIRMED)	**PUBLIC CAD / DESIGN** Full CAD + code (Onshape, GitHub)

LESSONS FOR STUDENTS

- The reveal that 'confirmed the dumper meta' -- a community thread analyzed what Limestone implied about REBUILT strategy.
- Localization + a custom driver dashboard show software/driver UX as a force multiplier.
- Pragmatic 'wood in the ball path' -- strong teams still use simple, cheap fixes.
- Fully open CAD + code makes this the single best reproducible study target of the season.

CAD release: https://www.chiefdelphi.com/t/1678-citrus-circuits-2026-cad-and-robot-code-release/521535 · Reveal: Limestone · Onshape CAD · Behind the Bumpers

#16 **6369 Mercenary Robotics** (full-ramp intake bot) — McKinney, TX — **EPA 274** — 77-9

Rebuild from a TURRET to a wide aluminum DRUM shooter; slapdown kicker-ramp intake -> rolling floor -> drum; net hopper (~78 fuel). Won Amarillo/Waco/San Antonio + Impact + EI [BtB transcript]

SHOOTER 3in aluminum DRUM (only ~1/6in wide -> too little momentum, so they added a 1.5 lb WCP STAINLESS flywheel at 1:3 - stainless beat brass, which wore down); drum on 4x Kraken X60 at 1:1; shooter geared 60:1 on a single X44 (down from 400:1); transfer = two 2in + one 1.25in rollers tilted back to passively funnel; back rollers two 1.25in silicone; ~20 ft/s for 60-70 balls; all motors packaged under the drum. Chose drum over turret for higher throughput + simpler/more-reliable software [BtB transcript]	**INTAKE** Slapdown intake (kept from robot 1) with a full KICKER RAMP (vs a kicker bar) for max fuel contact time + sustained compression (Citrus-style push into the indexer); 3 bars lift fuel high so the rolling floor can empty the hopper without much agitation [BtB transcript]
INDEXER / HOPPER Rolling floor of custom 3D-printed stub rollers linked by belts (the intake jackshaft is hidden inside a roller to use the full width; one roller pivots with the intake); hopper extends exactly 12in + a net -> ~50-55 fuel un-expanded, up to ~78 at 30in with the net up (for bump-overfill cycles) [BtB transcript]	**DRIVETRAIN** Swerve - swapped WCP X2 -> SDS MK5N (easy to replace), geared for speed + torque [BtB transcript]
ENDGAME Climber not detailed (turret -> drum rebuild).	**AUTONOMOUS** Auto routines not detailed; chose drum for simpler, more-reliable software.
VISION Three Limelight 4s (two front, one back) giving full-field localization in teleop + auto; one-button point-and-drive aiming with X-lock against defense; auto-aim auto-adjusts during autos [BtB transcript]	**PUBLIC CAD / DESIGN** Code on GitHub; no confirmed 2026 CAD

LESSONS FOR STUDENTS

- Best win-rate in the cohort (77-9) despite a lower EPA ceiling -- a case study in consistency and alliance leadership over raw scoring.
- 'Full ramp intake' points to a wide, forgiving ground pickup optimized for reliability.

Code (GitHub): https://github.com/mercs6369 · Behind the Bumpers · GitHub · DDS robot-pics-2026

#17 (study pick) **6800 Valor** Downpour — Austin, TX — **EPA 268** — 62-10

Elevator-launched fuel scorer with assisted-aim and bidirectional shooting

SHOOTER Dual carbon-fiber arm launcher fed up a ~47" elevator (~0.3 s travel); can shoot in two directions. Because the chassis is square, spacers pull the shooter plate inboard of the nut strips so the shooter could be made 4 fuel wide within the frame; the team notes how close the pulleys sit to the edge of the robot. Per the team in the release thread.	**INTAKE** Rotating intake synchronized with the arm for hand-off
INDEXER / HOPPER Internal feeder routing fuel intake→launcher; designed for fast maintenance access	**DRIVETRAIN** Swerve
ENDGAME Climber not detailed in the release thread; unconfirmed.	**AUTONOMOUS** Assisted-aim, bidirectional elevator-launched scorer; auto routines not detailed. A second robot, the Manor version of Downpour that the team calls Drizzle, has its own CAD branch in the same Onshape document.
VISION Top-mounted camera for assisted scoring/alignment with a reverse-kinematic software model (published in their tech packet)	**PUBLIC CAD / DESIGN** Public CAD (Onshape)

LESSONS FOR STUDENTS

- One living CAD assembly iterated all season (branches per version) keeps design history in one place.
- Maintenance-driven layout (top electronics, bottom battery plate, ziptie battery-lead hooks) -- design for the pit, not just the field.
- An elevator-launch architecture is a creative alternative to flywheel/turret shooters.
- Published a full Tech + Engineering-Ops binder -- strong documentation discipline.

Onshape CAD: https://cad.onshape.com/documents/15cd41462dc24ff986a4e549 · Tech binder (PDF) · CAD release thread · Reveal · Behind the Bumpers

#18 (study pick)

3467 Windham Windup (Open Alliance)

Windham, NH

EPA 266

75-10

High-throughput dumper with auto-aim shooter backup; software/autonomous-led

SHOOTER Primarily a bulk dumper (fill hopper, empty during on-shifts) with an auto-aim shot vs tower/trench	**INTAKE** Linear-slide retracting intake with polycarbonate hopper walls; tested 1- vs 2-motor intake
INDEXER / HOPPER Large hopper + indexer/tower rollers; tuned indexer floor angle and speed for throughput	**DRIVETRAIN** Swerve on Thrifty Narrow modules (switched from MK4i), CANivore/CANcoder, Choreo paths
ENDGAME Climber not detailed.	**AUTONOMOUS** Software/autonomous-led: a bulk dumper with an auto-aim shooter backup.
VISION Custom NVIDIA Jetson + 4-camera setup (CONFIRMED); AprilTag localization	**PUBLIC CAD / DESIGN** Full public CAD (Onshape) + Open Alliance blog

LESSONS FOR STUDENTS

- Autonomous CONSISTENCY beats peak score -- they were passed at alliance selection for inconsistent autos, then fixed odometry (lowered Choreo friction coeff, restored drivetrain current limits).
- New-carpet and CAN-terminator gremlins caused real losses -- document environmental variables.
- Their Open Alliance blog is a model of transparent post-event self-critique.
- Deliberate 'simple week-1 bot, planned mid-season upgrade' scheduling.

Onshape CAD: https://cad.onshape.com/documents/7ef2c49c5bb48212a6b180ee · Open Alliance blog · CAD

#19

9128 ITKAN Robotics Triple Threat

Plano, TX

EPA 265

60-7

Trench robot: cycloidal slapdown intake -> duct-tape conveyor -> transfer -> drum shooter, with a passive 'hopper flopper'. Won Space City 1 + Farmersville + an Impact award [BtB transcript]

SHOOTER Drum shooter: 2x Kraken X60 on a ~13 lb flywheel mass (steel flywheels + wheels); BANG-BANG control (from their FTC shooter experience) to ramp fast, an operator pre-ramp button while crossing the bump, and a VARIABLE transfer speed (slows/speeds vs the flywheel velocity-vs-setpoint so each ball takes max energy); ~25-30 BPS [BtB transcript]	**INTAKE** Flip-down SLAPDOWN intake on a CYCLOIDAL pivot drive (Kraken X60 -> cycloidals; chosen for ease of use + better ball compression than a linear intake); two rollers (3in hard wheels + a roller bar) on 2x Kraken X44; a pin-and-slot links the intake raise to pull the hopper back in (agitation) [BtB transcript]
INDEXER / HOPPER 7-roller conveyor with DUCT-TAPE flaps to fling balls into the transfer (the tape jumped BPS from ~18-20 to ~25-30); transfer = three 1in rollers + bottom stationary tubes for compression, rollers on slots for tuning. Passive 'hopper flopper' - a flexible hopper that collapses to go under the trench and re-expands to store ~80 fuel [BtB transcript]	**DRIVETRAIN** Swerve (not detailed); built as a trench robot [BtB transcript]
ENDGAME Climber not detailed.	**AUTONOMOUS** Drum shooter; auto routines not detailed (Impact award).
VISION A single Limelight facing the shooting direction; planned a 2nd but used software instead - an auto 'point-towards' the opposing hub to update odometry before crossing the bump (so it doesn't hit the trench) [BtB transcript]	**PUBLIC CAD / DESIGN** No public CAD (reveal + Behind the Bumpers)

LESSONS FOR STUDENTS

- A drum (revolver-style) shooter trades flywheel spin-up for high BURST rate: suited to empty-the-hopper volleys.
- Distinguish BURST throughput (20–25 BPS) from SUSTAINED (15–16 BPS) when specifying a shooter; they are different targets.
- Running two near-identical robots (9128/10340) multiplies build & test learning.

Video: https://www.youtube.com/watch?v=5HtLpv4AtlE · Reveal 'Triple Threat' · Behind the Bumpers

#20 1796 RoboTigers Sprout

Long Island City, NY

EPA 263

48-7

'Car Wash' spindexer fuel scorer

SHOOTER Launcher fed from a spindexer (launch type unconfirmed)	**INTAKE** Intake pre-fills/'primes' the feeder while the outer spindexer stack turns slowly
INDEXER / HOPPER 'Car Wash' spindexer -- two independently driven stacks, each on its own Kraken; outer stack spins slow during intake to evenly pack the hopper	**DRIVETRAIN** Swerve (specifics unconfirmed)
ENDGAME Climber not detailed.	**AUTONOMOUS** Auto routines not detailed ('Car Wash' spindexer).
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (photos on request)

LESSONS FOR STUDENTS

- Independently driven indexer stages (separate motors) let you intake and prime simultaneously without jamming.
- Slow-spin-while-intaking evenly distributes fuel -- a simple control trick for hopper packing.
- Cited by peers as 'one of the best robots in the world' despite no public CAD.

CD thread: https://www.chiefdelphi.com/t/robotigers-team-1796-presents-sprout-2026/516769 · Reveal 'Sprout'

#21 9483 Overcharge Enigma

Istanbul, Türkiye

EPA 261

38-6

Minimalist, low-motor-count spindexer scorer

SHOOTER Launcher fed from a spindexer (type unconfirmed)	**INTAKE** Two top rollers + a bumper-level mechanism (deliberately simple, no obvious kickers/ramps)
INDEXER / HOPPER Spindexer 'bowl' with a central cone wrapped in cat-tongue grip tape to fling fuel outward to the feeder	**DRIVETRAIN** Swerve (specifics unconfirmed)
ENDGAME Climber not detailed.	**AUTONOMOUS** Minimalist, low-motor spindexer; auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (reveal video)

LESSONS FOR STUDENTS

- Reputation for 'doing things with fewer motors than anyone thinks possible' -- fewer actuators means fewer failure modes, lighter/cheaper robots.
- A grip-taped cone in a rotating bowl can index fuel with almost no dedicated hardware: minimal dedicated hardware.

Video: https://youtu.be/tOYFbRIJ8CA · Reveal 'Enigma'

#22 (study pick)	**6329 The Bucks' Wrath** Roman I / Roman II Bucksport, ME	**EPA 260** 61-15

REBUILD story: 'Roman I' (turret + low-compression spindexer, ~8 BPS) -> 'Roman II', a WIDE aluminum DRUM shooter at ~23 BPS with a net hopper (~60-70 fuel) [BtB transcript]

SHOOTER Roman II: wide aluminum DRUM (custom 3D-printed segments over an ordered tube, CatTongue-taped) + an adjustable hood; flywheel speed and hood angle scale with distance (further = faster + hood up); ~23 BPS (vs ~8 on the Roman I turret). A safety routine auto-lowers the hood entering the trench, with a bigger safety zone the faster it crosses [BtB transcript]	**INTAKE** Custom silicone wheels (numbers printed) + an agitator linking all the rollers, CatTongue-taped for grip; a compact redesign over a wider v1 (a long bar that kept bending on hits was removed; flexible polycarb instead of Lexan to absorb impact); custom 3/8in-hex stub rollers instead of full-length shafts to save weight [BtB transcript]
INDEXER / HOPPER A net hopper raised capacity from ~35 to ~60-70 fuel; rollers + agitator feed CatTongue-taped rollers up into the drum (the whole top net/cover flips up for easy access to the battery, PDH, and motors) [BtB transcript]	**DRIVETRAIN** Swerve; built for serviceability (the top flips up to reach the battery / PDH / intake & swerve motors); prototyped on a wooden chassis to place electronics before final build [BtB transcript]
ENDGAME Climber not detailed.	**AUTONOMOUS** Vision-aimed turret; a safety routine auto-lowers the hood entering the trench (bigger zone the faster it crosses).
VISION THREE Limelights tracking AprilTags (added a 3rd on the shooter over the previous two side cameras for more accurate pose); pose-based controls - one button locks onto the target / passing mode, and an operator button auto-aligns even if pose drops [BtB transcript]	**PUBLIC CAD / DESIGN** Public CAD -- both versions (Onshape)

LESSONS FOR STUDENTS

- Zero-compression indexing (move fuel by geometry/rotation, don't squeeze it) reduces jams and wear.
- A simple 'upkicker' (static bar + side wheels) is a low-part-count hand-off from spindexer to turret.
- Released TWO full CAD versions (Roman I & II) -- a great study in in-season iteration.

Onshape CAD: https://cad.onshape.com/documents/014716f840fdbd882c1d691f · Reveal 'Roman' · Behind the Bumpers · Roman II CAD

#23 (study pick)	**2046 Bear Metal** Bearricade MkII Maple Valley, WA	**EPA 260** 78-16

'Big dumper' with a hopper-floor + drum shooter; simplicity-maximizing design. PNW District Champion, World Finalist [BtB transcript]

SHOOTER 4 lb, 4in-wide aluminum DRUM coated in CatTongue tape, ~3000 rpm; 24-25 BPS average (peaks 29-30), 4x Kraken X60; X44 hood with NO rollers (rollers hurt accuracy); hood -10deg to 60deg (60deg = bump-to-bump pass; -10deg shoots BACKWARD against the hub under defense while locking the wheels) [BtB transcript]	**INTAKE** Collector on 2x Kraken X60; two sets of CatTongue rollers + a 3rd 'pop-up' roller; stub axles for weight/durability; NO front beater bar - 1/2in polycarb + aluminum plates at contact points for resilience without the weight; an upper TPU-mounted protector roller keeps balls in the hopper [BtB transcript]
INDEXER / HOPPER Hopper extends 11.5in (kept under 12in so flex doesn't breach frame perimeter), double-fold polycarb top for rigidity + a net for capacity; ~70-80 fuel full, 50-60 while still fitting under the trench. Hopper bed = 11 rollers at a 21:20 ratio (back roller faster for grip), belts ~0.03in loose to cut current; floor pivots up for battery access. S-shaped (vs J-shaped) superstructure with a single 2-Kraken feed roller for throughput + less jamming + more hopper volume [BtB transcript]	**DRIVETRAIN** Swerve (AdvantageKit-style logging of swerve states + odometry) [BtB transcript]
ENDGAME Climber not detailed (big dumper, simplicity-max).	**AUTONOMOUS** Drive-and-empty dumper; auto routines not detailed.
VISION Vision pose updates fused with swerve odometry (logged); specific camera/coprocessor not named in the walkthrough (2025 code used PhotonVision + OV9782 AprilTag cameras) [BtB transcript]	**PUBLIC CAD / DESIGN** Bearricade Mk.II CAD (Google Drive)

LESSONS FOR STUDENTS

- 'MkII' naming reflects a deliberate v1→v2 redesign -- iterate on a known-good base rather than restarting.
- Known for fabrication quality and wire management -- clean execution scales to deep playoff runs.
- Pre-champs EPA peaked ~301, one of the highest all season -- a peak-performance benchmark.

Tech binder (Drive): https://drive.google.com/file/d/1heATIApMaUC-STLuVx7_C361IcmWv8U0/view?usp=sharing · CAD and Tech Binder release (CD) · Bearricade Mk.II CAD · Bearricade Mk.II Tech Binder · TBA 2026 · Behind the Bumpers

#24 (study pick)	**1987 Broncobots** Cyclone Lee's Summit, MO	**EPA 259** 54-6

'Cyclone': 360deg turret dye-rotor shooter fed by a linear deployable full-width intake; 21.75in tall to clear the trench; shoot-on-the-move [2026 Technical Book]

SHOOTER
Turret shooter kept to 4.25in height so the dye-rotor hopper keeps its volume. Flywheel = 4in WCP 60A urethane wheel on a dead axle with a custom integrated pulley hub, driven by two Kraken X60 at 1.25:1; the team states no flywheel mass was added, so RPM changes quickly for shoot-on-the-move. 'Floating hood' driven from the rear through a Thriftybot cycloidal gearbox on a Kraken X44 at 80.5:1 total, with a 10 lb constant-force spring that absorbs cycloidal backlash and a WCP through-bore encoder for true position; hood angle 0-90deg. Integrated 1in 45A hood rollers on a Kraken X60 at 28:20 reduce backspin. In the release thread the team states the floating hood was chosen over rack and pinion because a 4in rack limited hood angle. [2026 Technical Book; CD release thread reply]

INTAKE
Linear deployable full-width intake that collects up to 5 fuel at once, sized for a 30in effective intake width to take the 5 outermost neutral-zone columns in auto. 2-roller system moves fuel front-to-back. Polycarbonate structure with aluminium hub covers, rigid-mounted so impact loads go into the chassis, and large washers to spread roller loads. Rollers: 1.125in silicone and cat-tongue-wrapped polycarbonate on a Kraken X60 at 1.5:1; deployment on a Kraken X60 at 3.6:1. [2026 Technical Book]

INDEXER / HOPPER
Dye rotor, 22in diameter, mounted on its own plate for removal and service. 3D-printed diverter walls funnel fuel into the rotor; a wedge-shaped hook reduces the top footprint and allows reverse rotation to clear jams. Printed in Polymaker Fiberon PA612-ESD to prevent static buildup and reinforced with aluminium rods that carry turret load. Gearbox is coaxially driven with #25 chain on two X-contact bearings with fully enclosed bevel gears: rotation Kraken X60 at 20.67:1, feed wheels Kraken X60 at 1:1. Stated figures: 50 fuel hopper capacity, 8 fuel per rotor ring, 2.5 rotor rot/sec, 95 rot/sec interior wheels, 18 fuel/sec feed rate. Turret structure is sheet metal supported from the corner by a 1.5x1.5in tube that also carries the wires, reinforced with a carbon-fibre round tube; turret chain-driven by a Kraken X60 at 25.6:1 with a matched-reduction gearbox so the encoder reads absolute position. [2026 Technical Book]

DRIVETRAIN
Four West Coast Products X2i swerve modules with molded tread. Custom 3/8in billet frame rails place the battery nearer the frame edge and fully support the dye-rotor centre column. An electronics brainpan carries most wiring on its underside, lowering the dye rotor while keeping access. 0.125in polycarbonate skid plate protects the electronics. Single-piece bumpers backed by aluminium sheet metal include integrated hopper walls. [2026 Technical Book]

ENDGAME
The 2026 technical book describes four subsystems (drive train, intake, hopper and dye rotor, shooter) and does not describe a climber or other endgame mechanism.

AUTONOMOUS
The technical book states B-Line autonomous routines that dynamically recover from bumps and impacts from other robots. Intake speed scales with robot-relative velocity to conserve power. Specific path counts and scoring sweeps are not stated; read them from the released code.

VISION
Limelight 4 powered localization state with dynamic field zoning for automation; the Behind the Bumpers interview describes three Limelights for real-time pose. Shoot-on-the-move uses velocity control with augmented translational correction, gyro latency compensation, dynamic separation of fuel exit-velocity components for distance compensation through variable hood angles, and empirically measured fuel time-of-flight data for vector prediction. Real-time swerve wheel-slip detection corrects odometry. Code is Java command-based with a superstructure state machine and a Signal Fetcher that consolidates CAN interaction. [2026 Technical Book; BtB transcript]

PUBLIC CAD / DESIGN
Onshape CAD + GitHub code + 14-page technical book

LESSONS FOR STUDENTS

- Turret wiring: the umbilical runs from below the brainpan, across a tensioner, and up an aluminium tube into the turret, so all harness management sits under the robot. In the release thread the team lists what worked and what did not: a UHMWPE rope inside the harness takes tension off the wires, braided sleeving failed at the redirects and was replaced with automotive fabric heat-shrink, talc inside and Sailkote outside cut abrasion and friction, and fixed 3D-printed plus PTFE-tape redirects were used because pulleys on bearings would not meet the bend radius.
- Shooter height was set by the largest component: the team states that because the flywheel is 4in, the shooter was made 4.25in tall, which returned volume to the dye-rotor hopper and raised the turret structure enough to stop fuel jamming between the rotor hook and the structure.
- The floating hood was chosen over rack and pinion because a 4in rack limited hood angle. The Thriftybot cycloidal gearbox gave the reduction in a small package but had more backlash than expected, which a 10 lb constant-force spring absorbs.

- The technical book states measurable design targets before mechanisms: 15+ fuel/sec throughput, 30in effective intake width, 10+ fuel/sec shooter flow, sub-1s spin-up, and 21.75in height to clear the trench.

CAD release: https://www.chiefdelphi.com/t/1987-broncobots-2026-cad-code-resources-release/524348 · Reveal 'Cyclone' · Behind the Bumpers · 2026 CAD, Code & Resources release (CD) · 2026 CAD (Onshape) · Robot code (GitHub) · 2026 Technical Book (PDF)

#25 **1768 Nashoba Robotics** Nightshift — Bolton, MA — **EPA 259** (70-17)

Fixed full-width high-throughput floor-loader; volume shooter, NO turret

SHOOTER Full-width hooded drum shooter, 4× Kraken X60; 1/16" aluminum drum surfaced with CatTongue tape; 3D-printed hood segments between 1/16" plates; separate Kraken X60 entry roller	**INTAKE** 4-bar intake (roller plates, side hopper plates, upper links in SRPP); static kicker bars + 2× Kraken X60 rollers; deploy off a MaxSpline shaft under the floor
INDEXER / HOPPER Roller floor on WCP dead-axle stubs; belt shortened 0.030" to cut current; moving hopper sits under trench height; stretchy net gives ~65 fuel	**DRIVETRAIN** Swerve (modules unconfirmed)
ENDGAME Climber not detailed (fixed full-width, no turret).	**AUTONOMOUS** Volume shooter; auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (detailed reveal writeup); GitHub

LESSONS FOR STUDENTS

- You don't need a turret to be competitive -- a full-width fixed shooter + huge hopper wins on throughput.
- SRPP and dead-axle construction keep a wide superstructure light and stiff.
- Tiny tuning (0.030" belt shortening) measurably reduced current draw -- budget current deliberately.
- A compliant net hopper trades rigidity for capacity (~65 fuel).

CD thread: https://www.chiefdelphi.com/t/team-1768-nashoba-robotics-2026-robot-reveal-nightshift/519386 · Reveal 'Nightshift' · Behind the Bumpers

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#26 **8044 Denham Venom** Cat-5 — Denham Springs, LA — **EPA 257** (50-9)

Ran two robots: v1 (low 'bump'/under-trench) and v2 (a HighTide/Denham 'tide clone'); dye-rotor shooter [per 8044 team members on DDS]

SHOOTER Dye-rotor shooter, belt-driven (HighTide-style); v1 had belt skip only when jammed, v2 fixed it by tensioning the belt rather than adding 10DP gear stages [per 8044 on DDS - medium confidence]	**INTAKE** Unconfirmed (2025 robot noted for a clean intake-to-handoff path -- likely carryover)
INDEXER / HOPPER Dye-rotor fed (rotor flings fuel to the launcher) [per DDS photos]	**DRIVETRAIN** Swerve (specifics unconfirmed)
ENDGAME Climber not detailed (ran v1 under-trench + v2).	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** No public CAD (DDS robot-pics photos)

LESSONS FOR STUDENTS

- A 50-9 record with little public documentation: execution + driver practice can outrank publicity.
- Their history of mechanism handoffs is worth studying for reliable ball transfer.

DDS thread: https://discord.com/channels/1120162219502608426/1479491912305217637 · DDS robot-pics-2026 · TBA · DDS robot-pics-2026 thread (8044)

#27 (study pick)

1706 Ratchet Rockers Mirage

Wentzville, MO

EPA 256

45-2

Dual-turret shooter ('Mirage' - 'shooting so fast you think you're seeing double') fed by a single spindexer; Oklahoma Regional winner [BtB transcript]

SHOOTER TWO turrets, each with a front kicker + a back kicker; 6in aluminum flywheels (made by sponsor Custom Machine Works - tried steel, aluminum won); hoods rotate fully [BtB transcript]	**INTAKE** Full-width four-bar intake, 2x Kraken X60 (one for pitch, one for the rollers); 4 silicone rollers (3 carbon-fiber + 1 polycarbonate, all wrapped in white silicone) for grip; the intake doubles as a hopper extension to minimize moving parts [BtB transcript]
INDEXER / HOPPER A spindexer (re-iterated several times over the season) ramps fuel up into each shooter; kicker wheels fire the front + back kicker on each turret [BtB transcript]	**DRIVETRAIN** Swerve (odometry fused with vision in the pose estimator) [BtB transcript]
ENDGAME Telescoping-tube climber (Thrifty).	**AUTONOMOUS** Dual-turret, high fire-rate ('Mirage'); auto routines not detailed.
VISION 3x Limelight 4 + swerve odometry in the pose estimator; the Limelight hardware-manager Kalman filter rejects extraneous poses and only trusts MegaTag2 on good solutions; Limelights sit under each shooter + on the climber [BtB transcript]	**PUBLIC CAD / DESIGN** Onshape CAD (champs config 'Mirage') + code on GitHub

LESSONS FOR STUDENTS

- A 45-2 record is a case study in consistency over flash -- minimize failure modes, maximize cycle reliability.
- Watch their Behind the Bumpers for how a veteran team packages a clean, serviceable robot.
- Champs-config CAD is now public on Onshape -- dig into the dual-turret flywheel shooter and single-spindexer feed.

CD thread: https://www.chiefdelphi.com/t/522276 · CAD release (Mirage, champs) · Behind the Bumpers · Code (GitHub) · TBA 2026

#28

9496 LYNK Matterhorn

Spindale, NC

EPA 254

70-13

2-wide drum 'over-bumper reverse' shooter (modeled on 254's 2017 'Misfire')

SHOOTER 2-wide drum shooter, fixed 34deg hood (polycarb + foam, ~0.675in compression); 4 motors on one shaft (SDS brass + TTB-0106 flywheels, ~5.2 lb); flywheel RPM scaled by distance	**INTAKE** Roller floor with 'pulsing' (reverse-intake while shooting to empty the hopper)
INDEXER / HOPPER Roller floor: indexer rollers 0.375in, top floor roller 0.5in; indexer/floor/shooter run 1:1 at 12V	**DRIVETRAIN** MK5N-style swerve, ~26x26in, Spike grip wheels
ENDGAME Climber not detailed.	**AUTONOMOUS** 2-wide drum 'over-bumper reverse' shooter; auto routines not detailed.
VISION Single AprilTag camera for pose (primarily off the hub tags); a 2nd camera under consideration	**PUBLIC CAD / DESIGN** Open Alliance build log + GitHub

LESSONS FOR STUDENTS

- A young team (3rd year) reaching #28 EPA shows the value of an Open Alliance build log -- public iteration accelerates learning.
- Out-of-state district scheduling (VA→GA→SC) is a program-building lesson, not just a robot one.

CD thread: https://www.chiefdelphi.com/t/lynk-9496-build-log-2026/511797 · Open Alliance build log · Reveal 'Matterhorn'

#29 **5460 Strike Zone** Helios — Lapeer, MI — **EPA 252** — 57-11

Drum shooter mounted on a turret (a 'turret dumper'), ~30deg total turret travel

SHOOTER Drum shooter on a ~30deg-travel turret	**INTAKE** Unconfirmed
INDEXER / HOPPER Unconfirmed	**DRIVETRAIN** Swerve (historically SDS MK4i)
ENDGAME Climber not detailed (drum-on-turret 'turret dumper').	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** 2026 release pending (strong tech-binder history)

LESSONS FOR STUDENTS

- Strike Zone habitually publishes a full tech binder (2023, 2025) -- review those as a documentation template.
- A team known for 'accurate' shooters underscores investing in shot consistency over raw range.

CD thread: https://www.chiefdelphi.com/t/5460-strike-zone-2026-reveal/515689 · 2026 'Helios' reveal · 2025 CAD + binder

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#30 (study pick) **581 Blazing Bulldogs** Rubble — San Jose, CA — **EPA 251** — 47-19

'Rubble': dye-rotor turret shooter w/ 35-deg adjustable hood, pivoting floor intake, WCP X2i/X2c swerve

SHOOTER Dye-rotor turret ('floating turret' de-conflicts rotor table vs turret). Drum: 4 in OD x 1/16 in-wall aluminum wrapped in Cat Tongue grip tape, powered by four Kraken X60s. Adjustable hood with 35 deg of rotation on a Kraken X44 at 1:120 through a zombie roller; two 2 in Cat Tongue hood rollers geared up 1.64:1 plus three 1.5 in silicone-sleeved rollers on two X60s (1:1.5). Chose the short dye-rotor turret over a large dumper after week-1 Alpha Bot testing hit ~15 BPS but lacked consistency.	**INTAKE** Pivoting floor intake: pivots up/down for bellypan access and ball compaction; elastic X-pattern springing keeps constant tension so balls don't jam under the floor, and the front roller is lowered so balls can't cam under during the compaction sequence. Custom Onyx-printed pulleys with 1/16 in steel dowels to prevent layer shearing; installs onto the shooter tower with two bearing plates.
INDEXER / HOPPER Floor rollers feed the shooter tower. A CANrange measures hopper fill and auto-runs the rollers until a retroreflective sensor in the tower trips - buffering ~8 extra fuel in the tower without ejecting prematurely.	**DRIVETRAIN** 28.5 x 27 in frame (1/8 in-wall tube, 5/8 in corner chamfers to stay under the 110 in perimeter). Two WCP X2i + two X2c swerve modules, X1 gearing with 10T pinions, Kraken X60 drive / X44 steer; custom 1/16 in bottom plates to drive over the Depot, intake-side module top plates cut down for a lower intake.
ENDGAME Climber not detailed.	**AUTONOMOUS** Dye-rotor turret with adjustable hood (chosen over a dumper after Alpha-Bot testing); auto routines not detailed.
VISION 3x Limelight 4 (rear 'SHOOTER' + left + right), AprilTag MegaTag2; Limelight-only pipeline (no PhotonVision)	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape) + code (GitHub, team581/frc-2026)

LESSONS FOR STUDENTS

- Week-1 'Alpha Bot' archetype test: they built a quick dumper, measured ~15 BPS but poor consistency, and let DATA kill the archetype before committing - cheap insurance against a season-long mistake.
- Software power manager: a separate thread re-allocates current limits across subsystems in real time (e.g. prioritize the shooter while scoring) to prevent brownouts over a full match.
- Defensive autonomy: if all cameras drop, the robot tells the drivers and falls back to pre-set shots; dashboard toggles disable complex features mid-match.
- Turret wire management without e-chain: cable sleeving + surgical tubing for tension - low-cost and light.

Onshape CAD: https://cad.onshape.com/documents/766d37b113335d7a60cc2375/w/8b35fc383279c1954cec5b72/e/bd3188f47c23702bef9f1fc8 · CAD + Code release (CD) · Reveal (full Q&A) · Code (GitHub)

Additional well-documented robots for study (beyond the EPA top-30)

These rank below the EPA top-30 but each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. Ordered by EPA. ★ marks the recommended study picks.

#4 (study pick)	**2231 OnyxTronix** 2026 robot Israel	**EPA 312** 38-7

Swerve (4x SDS MK5n); 2x Kraken X60 turret shooter with a dynamic hood and Limelight 4; ground fuel intake; 3-belt conveyor + trigger; buddy climb. EPA is Local EPA (approx).

SHOOTER Flywheel shooter powered by 2x Kraken X60 (1.25:1). A dynamic hood/arc is driven by a Kraken X44 (58:1). The shooter sits on a 2x Kraken X60 turret azimuth (4.347:1) with continuous polar-velocity tracking around the HUB. A Limelight 4 sets hood exit angle and flywheel surface velocity from the robot's pose. Stated throughput ~12 fuel/s. All values per the Robot Catalog.	**INTAKE** Ground fuel intake. Rollers powered by 1x Kraken X60 (9.6:1); a 1x Kraken X44 (4.5:1) opens and closes the system linearly. The catalog lists full-width, multi-side, station and depot intake capability. Per the Robot Catalog.
INDEXER / HOPPER Conveyance in two parts. Conveyor: pulleys and timing belts driven by a Kraken X60 (3.125:1) plus a horizontal roller on the same motor (2.14:1); 3 conveyor belts keep ~3 fuel staged. Trigger: 3 synchronized rollers driven by a Kraken X60 (5:1), ~0.4 s intake-to-trigger. A passive, expandable 'cartridge' rides with the intake and holds up to ~50 fuel. Per the Robot Catalog.	**DRIVETRAIN** 4x SDS MK5n swerve modules: Kraken X60 drive (R1 ratio 7.03:1) and Kraken X44 steer (approx 26.09:1). Per the Robot Catalog.
ENDGAME Buddy Climb capability and a reliable endgame climb are listed as season goals. Climb mechanism specifics (motors, ratios) are not given in the catalog; unconfirmed.	**AUTONOMOUS** Won the Autonomous Award during the ISR district season. Specific routines are not detailed in the catalog; see the team's code.
VISION Limelight 4 on the shooter/turret for pose-based hood-angle and flywheel-velocity control and continuous turret tracking around the HUB. The catalog describes a custom pose/state filter with dynamic covariance scaling and line-of-sight shot suppression near the climb structure.	**PUBLIC CAD / DESIGN** Robot Catalog 2026 (PDF)

LESSONS FOR STUDENTS

- A public technical binder: nearly every subsystem lists its motors and gear ratios, so it is a turret-shooter packaging reference even though the CAD source files were not linked.
- Worked example of pose-based dynamic shooting: a Limelight 4 sets hood angle and flywheel velocity from robot pose, with a 2x Kraken X60 turret azimuth (4.347:1) tracking the HUB continuously.
- Conveyance is split into a 3-belt conveyor (keeps ~3 fuel staged) and a 3-roller synchronized trigger, achieving ~0.4 s intake-to-trigger and ~12 fuel/s.
- No downloadable CAD source files were linked in the release thread; the 36-page Robot Catalog is the study artifact. Rank 4 of 3705 (Local EPA approx), 39-6, ISR District Championship winner.

CD thread: https://www.chiefdelphi.com/uploads/short-url/sJlv0EIpVTo3jzyHGeTyMSwrbBX.pdf · CAD & Robot Catalog release (CD) · 2026 CAD (Google Drive) · Robot Catalog 2026 (PDF) · Statbotics 2231 (2026) · The Blue Alliance 2231 (2026)

#8 (study pick)	**1114 Simbotics** Simbot Tim Ontario, Canada	EPA 288 62-7

Wide DRUM dumper (deliberately NO turret); huge fuel capacity. Hall of Fame; won both Ontario districts + the Ontario provincial championship [BtB transcript]

SHOOTER Big belt-driven DRUM, an X44 + 3 more motors; hood uses only ~8deg of travel. They chose a drum (more versatile) OVER a turret (didn't suit their play style) and chose not to go under the trench [BtB transcript]	**INTAKE** 2x Kraken X60; balls hit a kicker bar + main roller; an 'agitating' intake with multiple states - it can vibrate fast to push balls into the indexer harder, or run gentler [BtB transcript]
INDEXER / HOPPER Extending hopper -> a moving 'rolling floor' (two rollers) carries balls up to the drum [BtB transcript]	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME A climber was designed and appears in the CAD, but per the team it was added on March 1 and removed by March 19 for weight, and stayed off the rest of the season as the game developed.	**AUTONOMOUS** All autonomous paths were built in the PathPlanner GUI, on full-field AprilTag localization. A simple PID-to-point with a timeout runs after crossing the bump to confirm the crossing before shooting starts. Autos are time-based: an auto that falls behind gives up and returns to the team zone, which the team says made pre-match auto coordination with alliance partners easier. Path count and scoring sweeps per auto are not stated [2026 code release].
VISION 3x Limelight cameras with MegaTag2 + AprilTags, fused with odometry for pose estimation [BtB transcript]	**PUBLIC CAD / DESIGN** 2026 CAD Release (Onshape)

LESSONS FOR STUDENTS

- The active-vs-passive roller split in the hopper is a concrete fix for game-piece bridging ('crystallizing'): one driven floor roller (Kraken X44 via a cross-robot belt) agitates the stack while the top rollers stay passive - a low-cost way to keep fire rate consistent.
- Intake was iterated over many versions with 1678: an example of cross-team prototyping focused on a single subsystem.
- A rank-8, 62-7 Hall-of-Fame robot with public CAD; a benchmark for overall packaging and mechanism integration.
- Endgame pragmatism: the climber was cut mid-season on a weight budget and never re-added - a reminder that a designed mechanism has to earn its weight.

Onshape CAD: https://cad.onshape.com/documents/7959d159451177786d52dd9a/w/535bc31b66b0cc442967f1d0/e/a263c1a625ed8cf2484ae82f · CAD Release (CD) · Code Release (CD) · Code (GitHub) · CAD (Onshape) · Statbotics 1114 (2026) · The Blue Alliance 1114 (2026)

#34 (study pick)	**5000 Hammerheads** Icarus Marblehead, MA	**EPA 236** 41-23

Turreted vision-tracking fuel scorer; full Open Alliance build + open-source code

SHOOTER Turret-mounted flywheel shooter with adjustable hood; camera-tuned auto-aim. Shot parameters (hood angle + wheel speed) come from InterpolatingTreeMap lookup tables with empirically-measured time-of-flight. Turret weight/reduction detail unconfirmed.	**INTAKE** Rack-and-pinion deploy driven by Kraken X44 motors (on TalonFX controllers), ~48:17 & 50:17 belt reductions; spin rollers
INDEXER / HOPPER Dye rotor plus a spin motor and dual feed motors (feed + follower); unjam logic with hook/feed stall detection	**DRIVETRAIN** Swerve (CTRE, Kraken motors on TalonFX controllers)
ENDGAME Climber not detailed.	**AUTONOMOUS** Shoot-on-the-move: 5 iterations of time-of-flight recursion with velocity- and latency-compensated turret tracking; robot auto-starts scoring ~2s before an active shift and stops ~1s after. Full autonomous routines are in the code / tech binder.
VISION 6x Arducam OV9281 on PhotonVision (front L/R/center + back L/R/center), MegaTag; welded-field AprilTag layout used at Worlds	**PUBLIC CAD / DESIGN** Championship CAD (Fusion 360) + open GitHub code

LESSONS FOR STUDENTS

- Well-documented shoot-on-the-move controls: InterpolatingTreeMap shot lookup tables, empirical time-of-flight, and velocity/latency-compensated turret tracking -- a concrete controls case study (see the Icarus 2.0 tech binder).
- Full Open Alliance build thread + public Fusion 360 CAD + open GitHub code (incl. a TurretIOTalonFX subsystem) document a turret robot end-to-end: a manageable scope for smaller teams.

Team site: https://a360.co/4wuUiK0 · Championship CAD (Fusion 360) · Tech binder (Icarus 2.0) · Open Alliance build thread · GitHub (2026Rebuilt)

#36 (study pick)	**359 Hawaiian Kids** Poi Pounder XXVII (V1 & V2) Hawaii, USA	**EPA 233** 69-11

Turret + adjustable-hood flywheel shooter, slapdown intake, two-stage tunnel+accelerator indexer, sliding hopper, swerve, climber; V1 & V2 CAD + 28-page binder

SHOOTER Adjustable-hood flywheel on a turret. Flywheel: 2x Kraken X60 (1:1) spinning three 4" 60A Stealth wheels + two trimmed SDS 4" brass flywheels (added moment of inertia), with five 1" Sushi accelerator rollers and a 3D-printed back ramp to compress the FUEL. Hood: Kraken X44 (167.5:1), 3D-printed sector-and-pinion, 28 deg range. Turret: 7.5" ID X-Contact bearing, custom 8ST/10DP gear, +/-200 deg (400 deg total) on a Kraken X44 (50:1), biased to the back-left corner for shoot-on-the-move.	**INTAKE** Motor-powered slapdown, three full-width horizontal rollers, touch-it-own-it. Deploy: Kraken X44 (50:1) via #25 chain on a coaxial shaft (also drives the floor belts) plus a spring-loaded kicker bar; the deployment motion also actuates the prismatic sliding hopper. Rollers: Kraken X60 (1.60:1), 1.5" polycarb tubes in silicone; 1/4" pocketed 7075 side plates, SRPP front guard. (V2 rebuild swapped this slapdown for a linear intake whose plates directly carry the extending hopper.)
INDEXER / HOPPER Two stages. Tunnel (Kraken X60, 1.88:1): top + bottom conveyors, two 15 mm belts each, align the FUEL into one controlled stream. Accelerator (Kraken X44, 1.46:1): three 2" Stealth + custom-lathed 2.5" Colson wheels fling the FUEL up into the turret. Fed by a sliding/extending hopper and a floor conveyor (back kicker roller -> Kraken X60, 6.30:1).	**DRIVETRAIN** SDS Mk5n R3 swerve (4 modules), Kraken X44 drive + steer, 26.09:1 steering, 4" molded spike-grip wheels (2.25" wide), 19.2 ft/s. 28x26" pocketed aluminum frame; Pigeon 2.0 IMU + CTRE CANcoders on all four modules.
ENDGAME Has a climber (per the 28-page binder); part of the V1->V2 design.	**AUTONOMOUS** Single-driver, two-button automation: auto-aim + auto spin-up by field position, hood auto-lowers under the trench; PathPlanner autos; 5-camera AprilTag localization; shoot-on-the-move.
VISION Five cameras feeding full-field AprilTag localization (robot pose, vision targets, and trajectories all modeled in physics simulation), logged via AdvantageKit / AdvantageScope 3D replay; specific camera brand not named in the binder. Drives single-driver, two-button stateless automation: auto spin-up + auto-aim to the hub or an alliance-zone corner by field position, with the hood auto-lowering under the trench; PathPlanner autos with a profiled point-to-point PID controller for precise climbs.	**PUBLIC CAD / DESIGN** CAD for both robots (Onshape) + 28-page technical binder

LESSONS FOR STUDENTS

- A two-stage indexer: a 'tunnel' that single-streams the FUEL, then an 'accelerator' that adds exit velocity. It decouples organizing balls from launching them.
- Stateless, two-button automation (one intake, one shoot) let them run a SINGLE driver: the robot auto-aims and auto-spins by field position and auto-lowers the hood under the trench.
- A mid-season rebuild (V1 slapdown + rail-carriage hopper -> V2 linear intake + directly-coupled hopper) is packaging-driven iteration; both versions' CAD are public for an A/B study.
- Validating every subsystem in physics-based simulation (FUEL, vision targets, and trajectories modeled) sped iteration and de-risked hardware before the robot existed.

CAD release: https://www.chiefdelphi.com/t/team-359-hawaiian-kids-2026-cad-release-q-a/521575 · CD CAD release + Q&A · Statbotics · The Blue Alliance

#32 (study pick)	**694 StuyPulse** HAL New York, USA	**EPA 229** 46-13

Swerve (WCP x2i) turret-shooter: hooded single-flywheel, ground intake, vertically-expanding hopper, L1 side-hang climber.

SHOOTER Hooded single-flywheel shooter mounted on a turret. Pre-season prototyping compared double-flywheel and single-flywheel overhand/underhand shooters; they built a hooded single-flywheel plate set with a polycarb sheet curved around the plates and a slot to adjust top-roller compression. The released Onshape model ('HAL 2026') has a 'Shooter + Turret' subassembly (33 features); wheel type, motors, and ratios are in the CAD, not stated verbatim in the thread.	**INTAKE** Ground intake; the team reused their 2025 coral ground intake for early testing and planned a wider intake to enlarge the hopper footprint. CAD has 'Intake' and 'Conveyor' subassemblies. Roller geometry/motors: see the CAD.
INDEXER / HOPPER Conveyor feeds fuel from the intake to the turret; the CAD 'HAL 2026' has 'Indexer' (8) and 'Conveyor' (4) features. The team's design goal was to intake and shoot simultaneously via the turret. Detailed geometry lives in the CAD.	**DRIVETRAIN** Swerve on WCP x2i modules, raised using the module's top-mount style (tubing sandwiched above the main plate with a WCP top plate) to gain ~1in clearance over the field bump; Kraken drive motors. The alpha/Kitbot used older MK4i modules from past robots.
ENDGAME L1 side-hang climber (CAD 'Climber', 8 features). The team deprioritized climbing versus hopper capacity but kept an L1 climb to preserve the endgame ranking point; an L3 climb was judged worth only a fraction of a cycle in this game.	**AUTONOMOUS** Not detailed in the thread; the turret was chosen partly for more time-efficient autos (shoot while intaking regardless of heading). See the code repo.
VISION Limelight (the CAD includes 'Limelight Views'; the team's plan stated adding a Limelight). Pipeline details unconfirmed; see the CAD/config.	**PUBLIC CAD / DESIGN** 2026 CAD (Onshape) - HAL

LESSONS FOR STUDENTS

- Turret trade study: 694 documented the pros/cons of a turret (shoot and ferry independent of drivetrain heading, keep the hopper full while intaking) against the added complexity and wire management, and chose to build one - a clear worked example of the decision.
- Vertically self-expanding hopper: the lid stays below trench height (about 22in) when empty and passively rises toward the 30in height limit as fuel fills the robot, adding a layer of storage without blocking trench access - a specific packaging idea worth studying.
- WCP x2i top-mount raise: mounting the tube above the module's main plate with a WCP top plate raised the whole chassis for about 1in of bump clearance without lifting the frame off the floor.
- Climb cost-benefit: they treated the L1 climb as a ranking-point insurance policy rather than a scoring priority after estimating the climb was worth only a fraction of a cycle.

Onshape CAD: https://cad.onshape.com/documents/2b44ae58b051bc8b70f03c29 · Open Alliance build thread (CD) · 2026 CAD - HAL (Onshape) · Code (GitHub, StuyPlus-2026) · Statbotics 694 (2026) · The Blue Alliance 694 (2026)

#36 (study pick)

1833 Team BEAN Clover

Georgia, USA (Peachtree)

EPA 225

60-8

Laned-shooter robot with a kicker-plate feed and a floor intake; full Onshape CAD + public code released. Drivetrain type not stated in the release thread.

SHOOTER Laned shooter (the team ran laned shooters rather than a drum). Before Worlds they weighed rebuilding from laned shooters to a drum and decided to keep the lanes, citing time, budget, and a judgment that other subsystems and driver practice would yield more. Wheel type, hood geometry, motors and gear ratios are not stated in the thread; read them from the Onshape CAD / code.	**INTAKE** Floor intake, reworked before Worlds. The team states that prior to Champs they were unable to intake from the depot, so they invested in the intake ahead of Newton. Roller count/geometry and motors are not described in the thread; inspect the Onshape CAD.
INDEXER / HOPPER Kicker-plate feed moves fuel to the shooter. The team notes they had issues with the strength of the kicker plates before Worlds. Detailed geometry is not in the thread; see the CAD.	**DRIVETRAIN** Not stated in the release thread. Drivetrain unconfirmed here (do not assume); see the Onshape CAD.
ENDGAME Not described in the release thread; see the CAD/code.	**AUTONOMOUS** Won the Autonomous Award (Google) at both Gainesville and the Peachtree District Championship. Specific routines are not detailed in the thread; see the code repo.
VISION Not mentioned in the release thread; unconfirmed. Check the CAD/code.	**PUBLIC CAD / DESIGN** CAD (Onshape)

LESSONS FOR STUDENTS

- A documented case of NOT rebuilding: 1833 weighed switching from laned shooters to a drum before Worlds and instead spent the time on intake and driver practice. A useful example of scoping late-season changes to expected payoff.
- Intake was the limiting subsystem going into Champs (they could not intake from the depot) and was the focus of their pre-Worlds work.
- A local-rank-36, 60-8 district team and Newton Division (Worlds) Finalist with downloadable CAD + code; an overall packaging reference.
- Two-time Autonomous Award winner: the public code repo is worth reading for autonomous structure.

Onshape CAD: https://cad.onshape.com/documents/11b1fc83786708b81b063e5c/w/a7854643bff51f67974cba0b/e/82ecb4d9374be4387f3eecba · CAD + code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics 1833 (2026) · The Blue Alliance 1833 (2026)

#45 5907 CC Shambots Chirp

Michigan, USA (FiM)

EPA 224

48-19

Swerve with a four-motor flywheel shooter, a 10-40 deg adjustable hood driven from a distance lookup table, an extending roller intake and a three-stage indexer.

SHOOTER

Four flywheel motors on CAN 31-34, configured as two counter-rotating pairs, on a 1:1 drum ratio with a 2.125 in flywheel radius and a 3000 RPM ceiling. Velocity tolerance is 6 rotations per second and modelled flywheel efficiency is 0.9. The hood is a separate motor on CAN 35 through a 1/9 x 10/160 reduction, travelling 10 to 40 deg under Motion Magic with a 2 deg tolerance. Shot parameters come from an InterpolatingTreeMap with 17 breakpoints between 1.0 m and 5.0 m of hub distance, and a second table covers passing shots. A latency coefficient of 0.5 compensates the shot for robot motion. Motor models are not named in the code; TalonFX is the controller, so read the motor from the CAD bill of materials. In the release thread the team states the shooter side plates were pocketed by hand with no FEA and each lost more than half its starting weight, and that the shooter pivot uses an aluminium bushing machined in house instead of a hex bearing to cut friction.

INTAKE

Two roller motors on CAN 11 and 12 plus one actuator motor on CAN 13. The actuator extends the intake from 2.0 in retracted to 11.493 in, under Motion Magic with a 0.5 in position tolerance, and has two further named positions at 0.0 in and 6.0 in used for agitation. Roller speed is 85 rotations per second. Roller geometry, materials and compression are not stated in the code; read them from the CAD.

INDEXER / HOPPER

Three driven stages, each its own motor: a floor stage on CAN 21 at 80 rotations per second, a tunnel bottom stage on CAN 22 at 75, and a tunnel top stage on CAN 23 at 85. The intake actuator positions double as an agitation cycle, so there is no separate agitator mechanism in the released code.

DRIVETRAIN

Swerve generated by the 2026 CTRE Tuner X swerve project and run on Phoenix 6. Drive reduction 7.03125:1, steer reduction 26.0909:1, 2 in wheel radius, free speed 5.12 m/s at 12 V. Module centres sit at 11.125 in longitudinal and 10.625 in lateral, giving a 22.25 x 21.25 in module footprint. Steer feedback is a fused CANcoder, the IMU is a Pigeon 2, slip current is set at 120 A, drive supply limit 50 A and steer stator limit 60 A. The motor model is not named in the code.

ENDGAME

The released code contains no climber subsystem. The Statbotics breakdown attributes 67.3 of the team's 223.8 EPA to endgame fuel, so the endgame is scoring rather than a mechanism. Confirm against the CAD.

AUTONOMOUS

PathPlanner, with the robot configuration loaded from the GUI settings file. The AUTO_PATHS array in the public release is empty, so the routines themselves are not in the release. The team won the Autonomous Award at the FiM Woodhaven district event.

VISION

PhotonVision with two forward cameras, pv_FrontLeft and pv_FrontRight, mounted 13.758 in forward, 3.375 in either side of centre and 13.145 in high, pitched 20 deg and yawed 10 deg outboard. Each camera has a 16 ft trust cutoff. Measurements are screened by an ambiguity threshold of 0.4 averaged over 100 frames with a distance scalar of 2. The AprilTag layout loaded is the 2026 REBUILT welded field, so tag poses match the welded field only.

PUBLIC CAD / DESIGN

Onshape CAD (2026 robot Chirp)

LESSONS FOR STUDENTS

- A complete, readable example of hood-plus-flywheel shot scheduling: one interpolating table for hub shots, a second for passes, and a latency term so the solution holds while the robot moves.
- The intake actuator carries named agitation positions in addition to deployed and retracted. Reusing the deploy actuator to unjam avoids a dedicated agitator.
- Indexing is split into three independently tuned stages running at different speeds, which is a cheaper way to control ball spacing than one long driven path.
- Vision trust is set per camera by distance cutoff rather than globally, and measurements are screened on a running average of ambiguity instead of a single frame.
- The AprilTag layout is hard-coded to the welded field. A team copying this code for an AndyMark field has to change it; the field dimension comments in Constants list both.
- The release thread records a manufacturing decision with its reasoning: an in-house machined aluminium bushing at the shooter pivot in place of a hex bearing, chosen for friction and for in-house capability rather than part count.

Onshape CAD: https://cad.onshape.com/documents/4bcf78cab5066972a785c135 · 5907 CC Shambots 2026 CAD and Code Release (CD) · Onshape CAD · Code (GitHub) · Statbotics 5907 (2026) · The Blue Alliance 5907 (2026)

#48 (study pick)

604 Quixilver Toploader

San Jose, California, USA

EPA 222

47-16

Swerve; single-stream turret shooter fed by a rotary 'dye rotor' serializer; polycarbonate intake arms; sliding corrugated-plastic hopper.

SHOOTER

Single-stream turret. The team chose a turret over a full-width launcher to score and feed on the move, accepting a balls-per-second reduction: a prototype serializer showed about 15 BPS single-stream against about 23 BPS full-width. The turret is packaged to minimise rotationally-swept volume, which preserves hopper capacity and ball flow into the bucket. The turret slider bearing is a stack of polycarbonate and aluminium plates; 3D printed PETG spacers between the ring and gear set bearing clearance by trial and error without remaking major parts, and the joint is heavily greased. Launch parameters come from an offline Monte-Carlo sweep of angle and velocity pairs, saved to a CSV table, selected for least sensitivity to error; launch vectors are then adjusted to cancel the robot's field-plane velocity for shooting on the fly. Motors, gear ratios and wheel specifications are not stated in the release post; read them from the Onshape CAD.

INTAKE

Independently driven intake arms, one Kraken X44 per side to deploy and retract, which fixed a chain-breaking problem in the mechanically coupled 'slap down' intake on their 2024 robot; the arms can flex endstop to endstop without damage. Motors stay inside the frame perimeter and drive the rollers through belts, with 3D printed TPU 'reverse pants' on each arm holding the belt up under deflection to prevent derailing. Arms are polycarbonate rather than SRPP, chosen to avoid permanent deformation that would change belt centre-to-centre distance and lose tension. Dead-axle rollers: polycarbonate rollers covered in silicone tubing, TPU pulley plugs, an individual dead axle per plug. A single-deploy kicker bar is spring-loaded with surgical tubing, held by TPU tube plugs and string across the assembly to avoid bolted stress concentrations.

INDEXER / HOPPER

'Dynamic Point of Capture' (DPC) serializer: a rotating mechanism, commonly called a dye rotor and inspired by paintball gun feeds, spins to scoop fuel out of a circular bucket, bringing the capture point to the fuel rather than the reverse. Rotation and up-feed run on independent motors so the ratio between them can be tuned in software. Omni wheels on the side feed rollers reduce forces and improve efficiency. Fuel is guided to the turret through a tube formed from corrugated plastic with 3D printed ends, each ball pushing the previous one up into the turret. The plywood bucket is split into two segments for battery and electronics access; plywood was chosen for stiffness-to-weight. The sliding hopper is two laminated layers of corrugated plastic, deployed by the intake and retracted by surgical tubing, with 0.090 in polycarbonate sandwiched around the slots; panels can be cut with scissors and are joined with tape.

DRIVETRAIN

Swerve. 27 in square chassis, sized to maximise the area available for the circular serializer bucket without custom modules or chamfered corners. Internal cross-bars form isolated battery and swerve compartments to protect electronics. Aluminium 'armour' integrated with the rear bumper mounting hardware protects the serializer from impacts inside the frame perimeter. Module make and gear ratios are not stated in the release post; unconfirmed.

ENDGAME

Not described in the release post; unconfirmed (see CAD).

AUTONOMOUS

Paths are pre-planned trajectories from Quikplan, the team's time-optimal swerve trajectory planner. Paths are designed around bump-only limitations and alliance-partner compatibility, using strategic pauses to avoid collisions and exploit gaps in traffic. Key path features including pauses can be edited up to the last minute without redeploying code. Alliance-partner match video is timestamped and compared against robot simulation to check compatibility. Won the Autonomous Award in the Newton Division.

VISION

PhotonVision MultiTag, four cameras (one on each side, two on the back). Each camera generates a robot pose; the software picks the pose from the camera seeing the most hub AprilTags, then the most total tags, and discards poses with nonzero ambiguity or an old timestamp. The turret is only considered ready to launch when velocity, launch angle and yaw are in tolerance and the cameras have a recent image of at least two tags.

PUBLIC CAD / DESIGN

Onshape CAD

LESSONS FOR STUDENTS

- The team identified serialization, not the launcher or the intake, as the limiting factor on balls per second, and then designed every other subsystem around the serializer. The release states the trade explicitly: about 15 BPS single-stream with a turret against about 23 BPS full-width.
- A rotary 'dye rotor' serializer brings the capture point to the game piece instead of moving the game piece to a fixed capture point. Independent rotation and up-feed motors let the ratio between them be tuned in software rather than in hardware.
- Material selection is justified against a failure mode each time: polycarbonate intake arms over SRPP to avoid permanent deformation that would slacken belts; plywood bucket for stiffness-to-weight; corrugated plastic hopper for repairability with scissors and tape.
- Design for the shop that exists: the stated constraints were tubing jigs, Fabworks, 3D printing and minimal router access.

- Launch parameters are solved offline by Monte-Carlo simulation and stored in a CSV table, which keeps onboard computation low and finds the launch pairs least sensitive to error.
- Automation was scoped so that one driver runs the robot: single-button scoring, automatic target selection by field zone, and automatic serializer start and stop based on turret readiness.

Onshape CAD: https://cad.onshape.com/documents/0baaf96036e07596972c4d48/w/7236f4fc3d9cbd0caa07ab6c/e/e54d06e7bb840dd08bd9d927 · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Statbotics 604 (2026) · The Blue Alliance 604 (2026)

#55 **8214 Cyber Unicorn** — China — **EPA 217** (16-8)

Ultra-clean sheet-metal turret shooter; #1 EPA in China, championship build quality

SHOOTER Custom CNC-machined flywheel (bore geometry designed to be cut with a 2.5mm end mill); turret-mounted shooter	**INTAKE** Roller intake (motors unconfirmed)
INDEXER / HOPPER Indexer/hopper present (capacity not stated in thread)	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber not detailed in the thread.	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed (not detailed in CAD thread)	**PUBLIC CAD / DESIGN** Full CAD release (Onshape)

LESSONS FOR STUDENTS

- Design parts AROUND your tools (DFM): they shaped the flywheel bore so a 2.5mm end mill could cut it.
- Outsource sheet metal to a sponsor, machine the rest in-house - that division of labor builds a clean, precise robot fast.
- Tidy wire management and packaging correlate with reliability and top EPA - visual order is an engineering signal.

CAD release: https://www.chiefdelphi.com/t/frc-cyber-unicorn-8214-2026-champ-robot-cad-release/521310 · CD champ-robot CAD release · Statbotics

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#56 (study pick) **4481 Team Rembrandts** (Open Alliance) — Eindhoven, NL — **EPA 214** (37-15)

Precision turreted hooded-flywheel scorer; strong controls

SHOOTER Turret hooded flywheel - their 4TH shooter design: a static back wall (they tried twin rollers but found them unneeded for high BPS), Teflon tape to cut friction/backspin, a custom flywheel to hold RPM, and a switch to METAL gears (the 3D-printed ones damaged easily) for champs. (Underlying: dual NEO Vortex single axle, ~370deg travel, herringbone rack-pinion hood 15-45deg) [BtB transcript]	**INTAKE** PASSIVE LINEAR intake (slides in/out, doesn't fold): NO deploy motors - constant-force springs + gravity drop it at match start; can't retract mid-match (deemed unnecessary); impact-resistant since it can fold in on a front hit without breaking [BtB transcript]
INDEXER / HOPPER 'Washing machine' spindexer: spins fast; iterated to a LARGER outer chamber + inner cone (helps balls settle -> higher BPS) with the gearbox + two motors packaged INSIDE the cone; lowered the washing-machine walls to admit balls easier. Turret cable chain kept taut by an elastic band (shortest at the anchor, stretches as it spins - never snags) [BtB transcript]	**DRIVETRAIN** Swerve (not detailed in the walkthrough)
ENDGAME Climber not detailed.	**AUTONOMOUS** Precision turreted hooded-flywheel with vision auto-aim; strong controls.
VISION Strong vision system: FOUR cameras at different angles to approximate robot pose; heavy focus on power management (per-subsystem current limits + supply voltage) [BtB transcript]	**PUBLIC CAD / DESIGN** Public CAD (GrabCAD) + Open Alliance thread

LESSONS FOR STUDENTS

- Gear-ratio documentation (45.7:1 flywheel, 58.5:1 hood): a model for sizing reductions to control-vs-speed needs.
- A through-bore encoder on the hood enables true closed-loop angle control, not open-loop presets.
- Their turret cable-sleeve fix (lengthening it to go from ~270° to ~370°) -- design the wire path for full range from the start.
- Authored the widely-used '3DM' game-analysis document.

GrabCAD: https://grabcad.com/team.rembrandts.4481-1 · Open Alliance thread · GrabCAD · Behind the Bumpers (full transcript)

#62 **3476 Code Orange** Sandspit
Irvine, CA

EPA 212
54-9

'Sandspit': four-bar intake -> pleated-belt roller floor -> full-width DRUM shooter (NO turret). Flow-obsessed; ~15 BPS. Orange County District [BtB transcript]

SHOOTER Full-width DRUM shooter (~15 BPS) with two massive ~5in in-house-machined flywheels (steel compound, mass concentrated at the outer radius for max inertia-per-weight, cut on a Haas mini-mill over a multi-day, multi-tool process); active hood on an in-house 10DP gear, symmetrical with a shaft + a 1:1 through-bore encoder for shoot-and-move; 3D-printed hood backings [BtB transcript]	**INTAKE** Four-bar intake with active ball control (its linear push turns the intake into a ramp funneling fuel toward the shooter); a 'scotch yoke' makes the hopper box follow the intake up/down; custom-machined 6061 sprockets (stock shortage); pivot driven through a 1in round tube to both ends. Tried coaxial roller power (balls caught) -> switched to Citrus's double-motor intake (much better) [BtB transcript]
INDEXER / HOPPER A 2:1 gearbox drives the roller floor + the first feeder roller; 'pleated belts' (TPU-printed cleats) on 3D-printed CF crown rollers agitate + push balls linearly down the ramp (replacing two top rollers that let balls fall back); crown belts drive the whole floor without belts between every roller (less friction). An 'epoxy coupler' roller mount (epoxy + embedded 4-40 nuts) cuts ~50% of stub-roller weight + allows quick roller swaps [BtB transcript]	**DRIVETRAIN** Swerve (not detailed publicly)
ENDGAME Climber not detailed (focus is on ball flow).	**AUTONOMOUS** Auto routines not detailed.
VISION Switched from Limelight to PhotonVision: SIX cameras - four AprilTag (two forward, two back) + two rear OBJECT-DETECTION cameras that find the densest fuel; 1600x800 with a custom ML algorithm that crops each frame to the AprilTags so it computes only on them (keeps ~50 fps and a stable pose); two Rubik Pi 3 co-processors [BtB transcript]	**PUBLIC CAD / DESIGN** Reveal video only (no downloadable CAD)

LESSONS FOR STUDENTS

- Treat ball flow as THE design objective - anything a ball can sit on, wedge above, or spin on is the enemy.
- Dividers/lanes that look organized often trap balls; smooth continuous paths beat segmented ones.
- Make the feeder transparent and film with high-speed cameras to diagnose flow problems you can't see at speed.
- Use the intake/box as a compactor to push fuel into the feeder while shooting.

CD thread: https://www.chiefdelphi.com/t/team-3476-code-orange-2026-sandspit-robot-reveal/516875 · CD reveal + flow philosophy · Reveal video · Behind the Bumpers

#71 (study pick)	**195 The CyberKnights** Big Daddy V2 'Biggest Daddy' Southington, CT	**EPA 206** 45-21-1

Over-bumper slapdown intake -> moving floor -> uptake -> DRUM shooter (swapped from a 4-lane shooter mid-season); custom ROS2 software. New England DCMP [BtB transcript]

SHOOTER 4in TITANIUM drum + a 1:1 BRASS flywheel + two carbon-fiber back rollers at 2:1; ~25 BPS. Started as a 4-lane shooter (~15 BPS) but balls jammed on the lanes -> removed the lanes and went to the drum [BtB transcript]	**INTAKE** Over-bumper SLAPDOWN deployed off an X60 (planetary 9:1 + two 3:1 for a fast deploy, esp. in auto); 3 polycarb rollers - a bottom kicker (no silicone; it caught on the bump) + two silicone front rollers; one Kraken X60 ~3:1. Swapped from an earlier rack-and-pinion + 4-roller intake that stuck/broke on hits (the new one is lighter) [BtB transcript]
INDEXER / HOPPER Moving floor = WCP stub-roller system (carbon-fiber rollers with silicone) on one Kraken X60 (~1:1, pushes balls hard into the uptake); uptake = six 1in carbon-fiber silicone tubes on 2x Kraken X60 [BtB transcript]	**DRIVETRAIN** Swerve (auto-shoot puts the swerve in X-mode to resist defense) [BtB transcript]
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Custom CyberKraken power/CAN boards; auto routines not detailed.
VISION FOUR Arducams -> a Mac Mini running PhotonVision; if any camera sees 2 AprilTags it fuses with odometry for full field awareness. Runs ROS 2 (a node per subsystem, so one crash doesn't take down the robot) + Molex locking connectors; full match logging (all node messages + 4 camera feeds) for replay [BtB transcript]	**PUBLIC CAD / DESIGN** Full CAD + code + docs (Onshape + SolidWorks)

LESSONS FOR STUDENTS

- Diagnose and fix recurring failures: they cured belt-snap by swapping to chain at the highest-load point.
- Custom 'CyberKraken' power/CAN boards with Anderson connectors (chosen over Molex SL for shock resistance) - real electrical-design depth.
- Win by refining a proven architecture through continuous in-season optimization.
- Public ROS-based robot code is an advanced software study target.

CAD release: https://www.chiefdelphi.com/t/team-195-the-cyberknights-2026-cad-code-and-documentation-release/521390 · CD CAD/code/docs release · Onshape CAD · Behind the Bumpers · GitHub

#79 6328 Mechanical Advantage Darwin

Littleton, MA

EPA 204

51-13

Software-first team: reliable turret shooter built around AdvantageKit + Northstar vision

SHOOTER Turret shooter with adjustable 3D-printed (PLA+) hood; 4in WCP Fairlane main wheel + 2in solid-core flex top wheel on a live single shaft; optional brass flywheel for added inertia; hood gear routed from WCP SRPP	**INTAKE** Compliant-wheel roller intake (prototyped publicly) feeding a centering/serializing path
INDEXER / HOPPER Intake -> centering hopper -> tower uptake -> turret/flywheel; serialization handled in software so balls feed one at a time	**DRIVETRAIN** Swerve; .090/.063 5052 aluminum sheet with lightening tricks
ENDGAME Climber not detailed in the build thread.	**AUTONOMOUS** Software-first: AdvantageKit + Northstar AprilTag localization (Orange Pi); auto-aim and autonomous routines in the released code.
VISION 'Northstar' custom AprilTag pipeline on Orange Pi coprocessors; alpha camera mounted behind the shooter hood for turret-to-hub alignment; full AdvantageKit/AdvantageScope logging + sim	**PUBLIC CAD / DESIGN** Open + code (AdvantageKit); CAD links in build thread

LESSONS FOR STUDENTS

- AdvantageKit IO-layer architecture and deterministic input->output ordering - a gold-standard software study.
- Northstar AprilTag localization on Orange Pi is a buildable, open-source vision stack.
- Routing custom gear teeth (SRPP/10DP) for turret and hood - a manufacturing technique they openly teach.
- Live-axle shooter packaging and brass-flywheel inertia trade-offs.

Code (GitHub): https://github.com/Mechanical-Advantage/RobotCode2026Public · CD build thread · Reveal 'Darwin' · GitHub (public code)

#80 3255 SuperNURDs The Thing

California, USA

EPA 202

37-23

'The Thing': 1678/254-style 'big dumper' high-flywheel FUEL scorer (rebuilt to this archetype in week 2); steel-drum flywheel with aluminum kicker/hood rollers; fully state-machine control architecture

SHOOTER Big-dumper flywheel scorer. Deliberately does NOT use an inertia (mass) wheel; instead a 4 in steel drum serves as the flywheel, storing energy in the drum mass itself, with 2 in aluminum drums acting as the kicker/hood rollers that meter and launch FUEL. The team converted to this 1678/254 'big dumper' layout mid-season (week 2), trading their original design for the proven high-flywheel dumper.	**INTAKE** unconfirmed from the release post (image + summary only) - big-dumper archetype implies a wide floor/roller intake feeding the elevated dumper; see the GrabCAD assembly for the intake and feed geometry.
INDEXER / HOPPER unconfirmed from the post - inspect the GrabCAD model for the hopper/indexer path into the dumper flywheel.	**DRIVETRAIN** Swerve (module specifics unconfirmed in the post). Notable control feature: 'Datum' anti-defense hardstops that let the robot quickly align to the hub, brace against the trench, and resist being pushed.
ENDGAME	**AUTONOMOUS**
VISION 3x Limelight 4.	**PUBLIC CAD / DESIGN** CAD release (GrabCAD)

LESSONS FOR STUDENTS

- How a mid-season pivot to the 1678/254 'big dumper' archetype (in week 2) can change a season to a top-100 world finish.
- A steel-drum flywheel (mass in the drum) as an alternative to a dedicated inertia wheel for a high-dumper shooter.
- 'Datum' anti-defense alignment hardstops: mechanical features that snap the robot into a known pose at the hub / trench to speed alignment and resist defense.
- A fully state-machine-based control architecture for a scoring robot.

GrabCAD: https://grabcad.com/library/frc-2026-team-3255-the-supernurds-1 · CD - 2026 CAD Release (Team 3255) · GrabCAD - The SuperNURDs 2026 · Statbotics 3255 (2026) · The Blue Alliance 3255 (2026)

#87 (study pick)	1156 Under Control Tempest Brazil	EPA 198 48-12

'Tempest': high-capacity swerve turret shooter -- intake → hopper → spindexer → feeder → ±250° turret → adjustable-hood flywheel, plus a climber

SHOOTER Turret-mounted adjustable-hood flywheel. Hood 15–55° via rack (1× NEO 2.0, 15.37:1 MAX Planetary). Flywheel: 2× Kraken X60 (1.4:1) driving 2× brass inertia flywheels (~150 g ea) + 2× 4 in urethane (WCP 60A). Turret: 6.5 in X-contact bearing, 42.24:1 (96T/10T), 1× Kraken X44, ±250°. Shoot-on-the-move via calibrated interpolation maps (distance → hood angle / RPM / flight time), 20 iterations per cycle.	**INTAKE** 'Touch-it-own-it' ground intake. 8 mm 7075 towers + 1/2 in polycarb arms (10DP rack); front 2.25 in (AM 35A) + rear 1.25 in rollers driven by 2× NEO Vortex (2:1, ~33 ft/s). Rack expansion by 1× NEO Vortex (5:1) deploys in under 1 s; 45° polycarb ramp.
INDEXER / HOPPER Hopper → Spindexer → Feeder. Hopper: polycarbonate + carbon-fiber, expands ~12 in to hold up to ~55 FUELS. Spindexer: 3D-printed cone, 2× NEO Vortex (4:1), keeps pieces moving (anti-jam). Feeder: 1× Kraken X60 (2.2:1), 16 compliant rollers, multi-piece flow with no dead zones.	**DRIVETRAIN** Custom SDS MK4n swerve on a 28×26 in frame (2×1 in Al tube, 2 mm bellypan). L1+ (14.2 ft/s); Kraken X60 drive + Kraken X44 steer (field-oriented); machined hubs with TPU treads for grip.
ENDGAME Climber on the turret shooter (binder-documented).	**AUTONOMOUS** Shoot-on-the-move via calibrated interpolation maps (distance->hood/RPM/flight-time), 20 iterations per cycle; 4-camera AprilTag fusion.
VISION 4-camera fusion: 2× Limelight + 2× Arducam → MultiCameraPoseEstimation with AprilTag corrections feeding odometry. Turret angle solved by trig + Chinese Remainder Theorem on two absolute encoders for a 720° usable range.	**PUBLIC CAD / DESIGN** Full CAD (Onshape) + 32-page technical binder

LESSONS FOR STUDENTS

- A 3DM (data-driven) process drove the design: priority list + archetype decision matrix + functional requirements (e.g. <22.25 in to clear the trench, <21 kg modules for air travel).
- The binder documents real motor / ratio / material callouts for every subsystem -- a model for a complete technical binder.
- Chinese Remainder Theorem on two absolute encoders yields a 720° turret range without re-zeroing: a controls trick.
- Shoot-on-the-move uses experimentally-calibrated interpolation maps (not real-time physics) iterated 20×/cycle -- a practical approach to moving shots.

Onshape CAD: https://cad.onshape.com/documents/96bff94aa06158572d9954cd/w/156a28a4c439f1f7e5f3bf04/e/3a4d945dab8b9e666f1c061c · CD CAD + tech binder release · 32-page technical binder (Drive) · Statbotics

#93 (study pick)

5987 Galaxia (i.m.o. David Zohar) (2026 comp bot)

Petah Tikva, Israel

EPA 197

6-6

Turret-on-hood shooter fed by a spindexer; physics-model firing solutions

SHOOTER Turret-mounted flywheel + adjustable hood; firing solution (wheel speed, turret/hood angle) from a physics model (air resistance, Magnus force) instead of a lookup table; auto-opens hood after trench from pose/velocity; Kraken X44; SOTM in progress	**INTAKE** Floor intake on revised roller geometry tuned for rapid multi-ball pickup without jamming; used for center-field steals in auto
INDEXER / HOPPER Spindexer -> curved ramp -> gate/kicker wheels; HDPE hopper walls (replaced polycarb) to reduce sticking; counter-rotating mecanum agitator (paintball-hopper inspired) forces balls down; 2x CANrange sensors; ~55-ball goal	**DRIVETRAIN** WCP SwerveX2c; Pigeon 2.0 IMU; PathPlanner autos for neutral-zone steals
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Physics-model firing solution (wheel speed + turret/hood angle from an air-resistance & Magnus model); hood auto-opens after the trench from pose/velocity.
VISION PhotonVision on Orange Pi with AprilTags around the central target; local 3D pose algorithm	**PUBLIC CAD / DESIGN** Full Onshape CAD (Open Alliance)

LESSONS FOR STUDENTS

- A physics-based firing model beats a lookup table - re-tuning means changing a few parameters, not re-populating a table.
- Material choice fixes jams: swapping polycarb hopper walls to slipperier HDPE reduced sticking.
- Borrow cross-domain ideas - paintball counter-rotating agitators inspired their mecanum de-jam wheel.
- Program in simulation first so new students can write/tune code before the robot exists.

CD thread: https://www.chiefdelphi.com/t/frc-5987-galaxia-2026-build-thread-open-alliance/505340 · CD build thread · In-season subsystem post · GitHub

#96

5406 Celt-X CX26

Hamilton, ON

EPA 196

45-15

Strategy-first: wide touch-it floor intake + flow-through hopper + turret single-file shooter

SHOOTER Turret-mounted flywheel, single-file feed; ~8 balls/sec peak for ~2s then ~5 sustained; re-geared flywheel pulleys (15:30 -> 22:24) for higher speed; used motion-profile / current control (6328-inspired) to tame large startup PID error	**INTAKE** Wide touch-it-own-it floor intake (3 balls wide); ran ~50% speed to avoid ejecting balls over the hopper; also acts as a compactor
INDEXER / HOPPER Flow-through hopper (~45+ balls) with spindexer-style feed; throughput drops as far-from-spindexer balls settle slowly; adding agitators	**DRIVETRAIN** Swerve; emphasis on under-trench driveability + fast cycles vs double-wide shooters
ENDGAME No climber - their strategy analysis flagged climbing as low-value.	**AUTONOMOUS** Strategy-first; auto routines not detailed.
VISION Unconfirmed (AprilTag targeting expected; not detailed)	**PUBLIC CAD / DESIGN** Onshape CX26 assembly (Open Alliance)

LESSONS FOR STUDENTS

- Build an Alpha bot to surface integration problems early (melting sidewalls, chain backlash, hopper dead zones) before the comp design.
- Know your shooter's real throughput CURVE - peak BPS means little if it collapses once near-spindexer balls are gone.
- Flywheel PID from a dead stop oscillates; a motion profile / constant-current model tames startup error.
- Single-wide shooters lose a measured throughput race to double-wide; compensate with turret + driveability + faster cycles.

CD thread: https://www.chiefdelphi.com/t/celt-x-5406-2026-build-thread-open-alliance/511200 · CD build thread · Alpha-bot throughput analysis · GitHub

#100 (study pick)

971 Spartan Robotics Mixtape

Mountain View, CA

EPA 196

36-28

'Mixtape': double-turret flywheel shooter (two shooters + spindexer) w/ lead-screw adjustable hoods; swerve

SHOOTER DOUBLE turret - two flywheel shooters, each with an adjustable hood driven by a lead screw. The lead screw was chosen to minimize the turret's swept radius: it tucks into the empty 'sides' instead of making the shooter longer. One iteration let the screw creep upward and lock the hood down; fixed by tapping and bolting the bottom of the screw (fix not shown in CAD). Wheel/motor specifics: read from the Onshape model.	**INTAKE** Ground intake feeding the hopper/spindexer (specifics not stated in the CD thread - see CAD).
INDEXER / HOPPER Simple spindexer feeding both shooters. Team reasoning: at high BPS it is easier to index into TWO shooters than one; they considered reviving their famous 2017 dye rotor (similar ball-shooting game) but judged it too finicky after losing the mentor talent behind the 2017 design.	**DRIVETRAIN** Swerve (module and gearing not stated in the thread - see CAD).
ENDGAME Climber not detailed (first public CAD year).	**AUTONOMOUS** Double-turret architecture; 2026 code in Java (a break from the historic C++ stack); auto routines in the released code.
VISION Custom in-house vision pipeline (971 tradition); 2026 hardware unconfirmed.	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape) + code (GitHub, frc971/971-second-robot-2026)

LESSONS FOR STUDENTS

- A double turret converts a hard indexing problem into an easier one: two feed paths at moderate rate instead of one path at high rate.
- Lead-screw hood actuation as a packaging move - use the dead space beside the turret to shrink swept radius rather than growing the shooter envelope.
- Institutional knowledge is a real design constraint: they rejected their own well-known 2017 dye rotor because the people who understood it were gone.
- First year of public 971 CAD, and 2026 code in Java (a break from their historic in-house C++ stack) - a newly study-able, established program.

Onshape CAD: https://frc971.onshape.com/documents/cabaa0c1c77517916df80783/w/48cb057db38a03cc202ff44e/e/96844befd4591dac162d657b · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

#103 (study pick)

341 Miss Daisy Miss Daisy XXIV

Ambler, PA

EPA 194

52-9

Hopperless 360+ turret shooter with a 4-ball-wide intake and a direct 4-2-1 ball path

SHOOTER Flywheel + fully adjustable hood (team's first) on a 360+ degree turret; fed straight from intake to shooter (no storage hopper). The turret is driven by a single 10DP gear at the centre of the turret gearing plate. Two 3D printed gears on the outer sides of that plate do not drive it: their tooth counts are coprime and each carries an absolute encoder, so turret rotation is read as a vernier sliding scale and multiple turns can be tracked. Per the team in the release thread.	**INTAKE** 4-ball-wide bottom roller intake feeding straight into the 4-2-1 path; there is no storage hopper to buffer it, so intake width is what sets the cycle rate. Motors and ratios are not stated in the release thread; read them from the CAD.
INDEXER / HOPPER No hopper - a '4-2-1' path narrows 4 lanes to 2 to 1 directly into the turret; ~8-ball working capacity	**DRIVETRAIN** WCP X2i swerve, 27in square base, 115 lb
ENDGAME Lightweight L1 climber kept deliberately cheap so mass/time goes to scoring.	**AUTONOMOUS** 360+ turret + adjustable hood for auto-aim; the hopperless 4-2-1 feed supports a fast continuous cycle.
VISION Single turret-mounted Limelight ('limelight-turret'), MegaTag2, IMU-fused; PhotonVision IO also present; ~8.25in turret-relative camera offset	**PUBLIC CAD / DESIGN** Full CAD + code + binder + pre-match checklists

LESSONS FOR STUDENTS

- Going 'hopperless' with a continuous 4-2-1 narrowing path removes a storage subsystem and its jam points - viable if intake-to-shoot rate is high.
- An adjustable hood lets one flywheel cover many distances; pair with turret+shooter code so aiming is automatic.
- Reliability is a process: they publish per-match pit checklists and run a role-specialized crew.
- Keep the climber cheap (lightweight L1) so mass/time goes to the scoring mechanisms that drive EPA.
- Multi-turn turret position is solved without a slip ring or a hard stop: two coprime encoder gears read as a vernier scale. It is a cheap, reproducible trick worth copying.
- Season result: 2x district event winner (Hatboro-Horsham and Centennial), undefeated at Centennial which was a team first, 154 district points before DCMP, 52-9.

CAD release: https://www.chiefdelphi.com/t/team-341-miss-daisy-2026-cad-code-and-documentation-release/521088 · CD CAD/code/binder release · Reveal (Miss Daisy XXIV) · Robot code (public)

#123 (study pick)

9072 TigerBots Asura + Sandstorm

Chesapeake, MD

EPA 186

52-18

Two-robot program: 'Asura' dual 720-deg twin-turret shooter, and 'Sandstorm' a 254/1678-style 4-wide dumper

SHOOTER Asura: TWO independent 720-degree turrets, each its own flywheel (~10 BPS combined). Sandstorm: 4-ball-wide top-level dumper/shooter (~19 BPS). Motor packaging iterated across test bots	**INTAKE** Full-width roller intake feeding the indexer; intake-deploy gearbox added last
INDEXER / HOPPER Asura: ~45-ball hopper. Sandstorm: ~75-ball hopper. Concluded early they'd be 'indexer-limited' and hunted the fastest indexer concepts	**DRIVETRAIN** CTRE swerve, Kraken X60 drive+steer, CANcoder, 7.67:1, 1.95in wheels, ~4.0 m/s
ENDGAME Climber not detailed (two-robot program).	**AUTONOMOUS** Concluded indexing throughput was the binding constraint; auto routines not detailed.
VISION 3x Limelight (left/center/right) via MegaTag	**PUBLIC CAD / DESIGN** CAD + code for BOTH robots (Onshape/GitHub)

LESSONS FOR STUDENTS

- Prototype the game piece FAST and find your binding constraint: they proved intake/shoot were easy and that INDEXING throughput was the real limit.
- Mine prior art - they studied 254 & 1678 plus RI3D/Open Alliance before committing.
- A dual-720-turret robot is buildable by a young team if you stage it (test bot first, then turret controls + tensioning).
- Same-team A/B study: directly compare their dual-turret vs dumper design choices.

CAD release: https://www.chiefdelphi.com/t/team-9072-tigerbots-2026-cad-and-code-release/521421 · CD CAD + code release (both bots) · Team robot page · Robot code (public)

#125 (study pick)

2767 Stryke Force (2026 comp bot)

Kalamazoo, MI

EPA 186

55-12

High-quality fabrication: single shoot-on-the-move turret + spindexer on a DIY swerve (82% win rate)

SHOOTER Single turret with a 'breadbox' adjustable hood; 3D-printed turret bearing; inverted motor mount; built for shoot-on-the-move	**INTAKE** Roller intake (hardened States->Worlds; no major redesign)
INDEXER / HOPPER Spindexer feeding the turret; intentionally flexes out of the way for battery changes (ergonomics); no drum	**DRIVETRAIN** DIY in-house 'Third Coast' swerve on a Chrome-Moly frame; many parts 3D printed
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Built for shoot-on-the-move (single 'breadbox'-hood turret); fast-iterated vision.
VISION FOUR cameras (two on each corner) on TWO Orange Pis (tucked under the batteries) running their custom 'Walleye' AprilTag software (in use since AprilTags launched); used heavily to correct the gyro/odometry in this odometry-driven game [BtB transcript]	**PUBLIC CAD / DESIGN** Onshape (2026 robot)

LESSONS FOR STUDENTS

- DIY swerve design - their full mechanical design doc (Third Coast) is public.
- 3D-printed turret bearing + inverted motor mount - clever, copyable packaging.
- A spindexer that flexes for serviceability - design for repair / battery access.
- Iterate for robustness (intake hardening) without scope creep.

Onshape CAD: https://cad.onshape.com/documents/a4cd67d32bba9211459890b2 · CD 2026 CAD · Onshape CAD · Third Coast swerve design doc · GitHub · Behind the Bumpers (full transcript)

#127 **360 The Revolution** Rainmaker XXVI
Tacoma, WA
EPA 186
40-20

Full-width turret shooter, bang-bang flywheel, shoot-on-move; 254/2910-style code

SHOOTER Full-width turret; 6328-style bang-bang flywheel wrapped in a state machine; shot interpolation from data; SOTM capped ~0.5 m/s (rising); time-of-flight tracking; passing/shooting auto-selected by robot position + match time	**INTAKE** Independent intake state machine (full-width feed); mechanical detail not fully specified
INDEXER / HOPPER Feeds the turret; managed within the superstructure state machine	**DRIVETRAIN** Swerve; PathPlanner autos (kept partly command-based to avoid state-machine conflicts)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Shoot-on-the-move (~0.5 m/s, rising) with time-of-flight tracking; passing vs shooting auto-selected by robot position; 254/2910-style state machine.
VISION Limelight + AprilTags (AprilTag field layout; a lazy-init bug once cost a loop overrun + ~5 hrs at field)	**PUBLIC CAD / DESIGN** Onshape CAD + public GitHub

LESSONS FOR STUDENTS

- Superstructure/state-machine control (a la 2910/254 ControlState) is more flexible & testable than scattered commands.
- Adopt proven patterns wholesale: 6328 bang-bang + AdvantageKit IO layer, 254 ControlState for isolating subsystems to test.
- A CI pipeline (build/format/sim gates) + an AI first-pass review scales code quality - caught ~35 bugs incl. a missing switch break that would have broken aiming.
- The real bottleneck is PR review / learning / robot test time, not code generation.

Onshape CAD: https://cad.onshape.com/documents/91500087e03b9e406a527d6d · CD build thread · Onshape CAD · GitHub RainMaker26

#158 (study pick) **190 Gompei and the HERD (WPI)** V1 'Doomspiral' + V2 'TURNOVER'
Worcester, MA
EPA 176
78-26

College-mentored team that built and CAD-released TWO distinct competition robots

SHOOTER V2 TURNOVER is a turret shooter with a belt 'kicker' that feeds/compresses the ball into the flywheel (they stayed turret while many moved to drums)	**INTAKE** Kraken X60 FOC roller intake (integrated 'turnover' architecture)
INDEXER / HOPPER Kraken X60 FOC rollers route fuel through the 'turnover' path into the turret	**DRIVETRAIN** CTRE swerve, Kraken X60 FOC
ENDGAME Climber not detailed (two robots: V1 'Doomspiral' + V2 'TURNOVER').	**AUTONOMOUS** Auto routines not detailed; the study value is the V1-vs-V2 A/B comparison.
VISION 3x Limelight 4 (shooter-mounted, climber, intake positions), MegaTag2 via their GompeiLib	**PUBLIC CAD / DESIGN** Onshape - both robots (Doomspiral V1 + TURNOVER V2)

LESSONS FOR STUDENTS

- Comparing two of one team's own robots (V1 vs V2) is a rare A/B design study - what changed and why.
- Turret-vs-drum trade study: they went turret while the field trended to drums - good design-decision reasoning.
- A belt-kicker that compresses the ball into the flywheel is a clean, copyable feeding concept.

Onshape CAD: https://frc190.onshape.com/documents/ba8a365713d40fc24cf0b54f · CD 190 CAD release (both robots) · TURNOVER showcase · Onshape (TURNOVER) · Robot code (2k26)

#153 (study pick)	7415 Jaguar Robotics California, USA	EPA 176 49-17

Swappable-hopper quad-lane shooter ('Noah's Ark'): tall ~100-fuel config or short trench config, swapped in <5 min

SHOOTER Fixed (non-turret) QUAD launcher - four lanes, each on its own Kraken X60, spin-up under 0.25 s; kept under 22 in tall to pass under the trench.	**INTAKE** 'Lintake' - a 5+ fuel-wide LINEAR intake with a powered kicker roller.
INDEXER / HOPPER Swappable hopper (~100-fuel tall config OR a short trench config, swapped in under 5 min via 8 bolts) feeding six powered floor rollers with zero dead zones.	**DRIVETRAIN** 23.5 x 31.5 in base on SDS MK5n swerve (geared to R2), 7075 punch tube.
ENDGAME Designed around scoring rather than climbing the Tower (climber not a focus).	**AUTONOMOUS** Full-field localization + shooter automation for scoring autos; PathPlanner-based routines (binder details the autonomous + shot-automation sections).
VISION 4x Limelight 4 (LL4) full-field localization.	**PUBLIC CAD / DESIGN** Full CAD (Onshape) + code (GitHub) + web technical binder

LESSONS FOR STUDENTS

- A swappable hopper that reconfigures tall-vs-trench between matches exploits the multi-configuration inspection rules - capacity when you can, low profile to clear the trench and beat defense.
- An 'Alphabot' pre-season testbed surfaced problems early (inaccurate drum shot, illegal start config) before the real build.
- Four independent shooter lanes (one Kraken X60 each) give a high fire rate with sub-0.25 s spin-up.

Onshape CAD: https://cad.onshape.com/documents/d7e50bc5f2f3accc3e502b3e/w/6c7c691839b17298be38e609/e/a406643d94e0f4b9284c165c · CAD/Code/Binder release (CD) · Technical Binder · CAD (Onshape) · Code (GitHub) · Statbotics

#213 4915 Spartronics ARTEMIS

Washington, USA (PNW)

EPA 159

45-23

Swerve with a turreted shooter and adjustable hood on a pivoting superstructure, a feeder and indexer pipeline, a climber, and three-camera pose fusion.

SHOOTER

Two motors, lead on CAN 22 and follower on CAN 23, with a sensor-to-mechanism ratio of 0.9375 and an 80 A supply limit that is disabled by default in the configuration. The auto-aim layer caps shooter speed at 120 rotations per second and treats the shooter as ready within 4 rotations per second of target. A modelled 10.695 percent loss is applied on shooter-to-ball transfer. The hood is a separate motor on CAN 21 through an 85.33:1 reduction with a 40 A limit and a 4 deg aim tolerance. Motor models are not named in the code.

INTAKE

Lead motor on CAN 15 through a 4:1 reduction with a 40 rotations per second ceiling. The autonomous constants give the intake a 213.05 mm extension beyond an 818.5 x 875.65 mm robot with 72.7 mm bumpers, which is the envelope the path planner respects. Roller count, material and compression are not stated in the code; read them from the CAD.

INDEXER / HOPPER

A two-element pipeline. The indexer is a single motor on CAN 17 through a 5:1 reduction, capped at 22 rotations per second with a 60 A limit. The feeder that follows it is a motor on CAN 18 through a 34:9 reduction, capped at 100 rotations per second with a 60 A limit, with a ball-detection sensor and a 0.5 s debounce before the superstructure treats the robot as empty. Pipeline state changes are rate-limited to 0.2 s.

DRIVETRAIN

YAGSL-configured swerve. Each module uses TalonFX controllers for drive and steer with a CANcoder azimuth encoder, and the IMU is a Pigeon 2, all on one named CAN bus called Hydra. Drive reduction 6.03:1 on a 4 in wheel, steer reduction 26.09:1, drive current limit 60 A and steer 40 A, with a wheel grip coefficient of 2.255. Module centres are 12.164 in forward and 9.586 in lateral. Odometry runs at 120 Hz. A slip detector watches for 2.2 rotations per second of wheel slip over 5 debounce cycles and swaps in degraded pose standard deviations for 0.3 s. Driver speed is governed to 15 percent of maximum and 20 percent angular while shooting at the hub, and 50 percent while ferrying. The motor model is not named in the code.

ENDGAME

Climber on CAN 24 through a 20:1 reduction with a 60 A limit, travelling -3.5 to 3.5 in the code's units, with velocity and acceleration limits of 10. The autonomous layer carries its own climb path constraints. Mechanism form is not stated in the code; read it from the CAD.

AUTONOMOUS

Not a fixed path library. The routines are built from named field points (tower, hub, outpost, depot, centre) with zone transitions, a pre-alignment step and a drive-to-POI command, run through a PPHolonomicDriveController with separate path constraints for drive-to-centre, bump, intake, climb and align moves. Tolerances are 1.5 cm, 3 deg and 2 in/s. The chooser composes routines rather than selecting pre-drawn paths.

VISION

Three PhotonVision cameras, named evan on the front tower pitched 30 deg down, val on the back tower pitched 30 deg down and facing rearward, and daniil mounted at the RIO pitched 26 deg down and yawed 297 deg. A Limelight turret camera called limelight-argos, with its own turret-to-camera transform, is present in the code but commented out of the primary camera list. Results pass a filter stage (90 ms maximum latency, 7 m maximum single-tag distance, 0.18 maximum ambiguity, area band 0.04 to 10) and then a fusion engine that requires at least two cameras inside a 20 ms timestamp window and rejects outliers beyond two sigma. Standard deviations are computed per frame from a 0.41 m base weighted by ambiguity 0.7, tag area 0.6 and latency 0.2, rather than fixed. A 0.5 s turret history buffer lets a turret-mounted camera be referenced back to the chassis. The AprilTag layout loaded is the 2026 REBUILT welded field.

PUBLIC CAD / DESIGN

Onshape CAD (2026 robot ARTEMIS)

LESSONS FOR STUDENTS

- The clearest multi-camera example in this set. Camera vendors sit behind one processor interface, so a Limelight and three PhotonVision cameras feed the same fusion engine and one can be commented out without touching the consumer.
- Vision standard deviations are computed per frame from ambiguity, tag area and latency instead of being a tuned constant, and the fusion engine refuses a pose unless two cameras agree within 20 ms.
- Slip detection is wired into the pose estimator: detected wheel slip swaps the odometry standard deviations for 0.3 s rather than disabling odometry.
- Driver speed governing is part of the shot solution. The drivetrain is held to 15 percent of maximum while aiming at the hub and 50 percent while ferrying, which is a software answer to a problem often solved in hardware.
- The turret is geared about 92:1 with the absolute encoder on the final 11:84 stage, and the superstructure enforces a turret window while the pivot sweeps, which is a worked example of interlocking two articulated axes.
- Autonomous is composed from field points and zone transitions rather than drawn paths, with the robot envelope and intake extension as explicit constants the planner respects.

Onshape CAD: https://cad.onshape.com/documents/16efabe4144263150bdc02ed · 4915 2026 CAD and Code Release (CD) · Onshape CAD · Code (GitHub) · Statbotics 4915 (2026) · The Blue Alliance 4915 (2026)

#224 (study pick)	**3847 Spectrum** Infrared (FM) Texas, USA	**EPA 158** 39-27

'Infrared': turreted hood launcher, extending roller-bed intake, ball-tower indexer; 5-robot dev line (AM/XM/PM/FM/Photon); swerve

SHOOTER Turret-mounted launcher with hood. Per the blog motor budget: turret on a Kraken X44, 2x Kraken X60 launcher flywheel, hood on an X44 or servos; Limelight rides on the turret (visible in the FM CAD next to a Talon FX-controlled motor).	**INTAKE** 12 in extending roller-bed intake (XM sketches): rack-gear extension (linear or slightly curved), single 2 in roller with a fold-down kicker bar, deploy motors stationary near the front frame rail. Final geometry sits 5 deg over the swerve module - the smallest angle that still compresses FUEL into the carpet (per mentor Q&A).
INDEXER / HOPPER Roller bed feeds two-ball inputs through 2-to-1 gates into a single ball tower that feeds the turret; lower powered rollers plus top rollers vector the balls. Hopper walls are laser-cut polypropylene (used as a laserable, Loctite-safe polycarb substitute). They also published a 4-lane 'sideways spindexer' concept (4x throughput idea) they considered for the rebuild.	**DRIVETRAIN** Swerve. AM (alpha) ran an existing MK4i base with MK5n considered for space claims; later machines kept the same footprint (final module not stated in the blog).
ENDGAME Climber not detailed in the released dev-line CAD.	**AUTONOMOUS** Dev-line approach (AM/XM/PM/FM): final vision/auto decisions deliberately deferred until after seeing the game at real events.
VISION Limelight mounted on the turret; blog discusses aiming via odometry vs turret-camera tx/ty, with AprilTags across the field providing pose updates.	**PUBLIC CAD / DESIGN** CAD for all 5 robots (Spectrum 2026 CAD index page)

LESSONS FOR STUDENTS

- Their AM/XM/PM/FM/Photon 5-robot cadence intentionally delays final design decisions until AFTER seeing the game played at real events - architecture as a season-long iteration, not a kickoff bet.
- Publishing every machine's CAD (alpha through final) gives a rare public record of how one architecture matures across four rebuilds in a single season.
- Small geometric insights matter: the intake angle is exactly the minimum (5 deg) that clears the swerve module while still pressing FUEL into the carpet.
- Material pragmatism: polypropylene hopper walls because they laser-cut safely (polycarb doesn't) and take Loctite - process constraints driving material choice.

CAD release: http://2026cad.spectrum3847.org/ · OA Build Blog (CD, 416 posts) · 2026 CAD - all 5 robots · Statbotics

#232 **1756 Argos** Miami
Illinois, USA

EPA 155
32-17

'Miami': MK5n swerve, Madtown(1323)-style roller intake v2, short spindexer indexer, rear Limelight mount; launcher detail in CAD

SHOOTER Launcher specifics not documented anywhere public - the top-rear roller launcher assembly is visible in the Fusion model; read ratios/motors from the CAD.	**INTAKE** 'Madtown_Intake_v2' (92-part subassembly): a roller ground intake in the style popularized by 1323 Madtown Madness, iterated to v146 in their Fusion history.
INDEXER / HOPPER 'Spindex_Short' - a compact spindexer, by far the largest subassembly in the model (397 parts). Combined with the intake naming, the architecture is intake -> short spindexer -> launcher.	**DRIVETRAIN** SDS MK5n swerve base ('Swerve_Base_mk5n_2026', 93 parts) on a sheet-metal chassis with riveted side panels.
ENDGAME Climber not documented in the public CAD.	**AUTONOMOUS** Rear Limelight mount modeled for vision coverage; auto routines not public.
VISION Limelight on a dedicated rear mount ('back_LimeLight_Mount'); two camera-FOV volumes ('FOV Off') are modeled in the assembly to plan vision coverage.	**PUBLIC CAD / DESIGN** Full robot CAD (Fusion 360 / A360)

LESSONS FOR STUDENTS

- Modeling camera FOV volumes directly in the robot assembly is a cheap way to verify vision coverage before building anything.
- Naming a subsystem after its inspiration ('Madtown intake') and iterating it 140+ versions shows healthy borrow-then-refine engineering.
- A 'short' spindexer keeps the proven anti-jam architecture while reclaiming vertical space - worth comparing against full-height spindexers like 1756's previous seasons or 3847's concept studies.
- A complete, well-organized Fusion 360 release is rare (most teams publish Onshape) - valuable for Fusion-based teams wanting a reference structure.

Team site: https://a360.co/4c8kxxw · Full CAD (Fusion/A360) · Spectrum CAD Collection · Statbotics

#252 **1714 MORE Robotics** Steel (rebuild of Flint)
Wisconsin, USA

EPA 150
43-20

'Steel' (mid-season rebuild of 'Flint'): 17 in-wide rear drum shooter with adjustable hood, 4x Kraken X44; roller-floor intake; deployable polycarb blocker; only the drivetrain and intake pivot carried over from Flint.

SHOOTER 17 in-wide rear-mounted drum shooter with an adjustable hood, powered by 4x Kraken X44. The team chose X44 over X60 for power density (W/lb) because making weight drove the rebuild: they budgeted ~200 A for a shot (~120 A to the shooter, ~80 A to rollers/indexer), and X44/X60 give similar peak power at a 40 A limit while four X44s save ~2 lb. The earlier robot 'Flint' used a single-wide FRONT fixed shooter (2x Kraken X60, later 2x X44) with an adjustable hood. Drum diameter, wheel/hood geometry and gear ratios beyond these are not stated in the thread - read them from the Onshape model.	**INTAKE** Roller-floor intake; the intake pivot was one of only two subsystems carried over from Flint. Many rollers/pulleys are 3D printed (eSun PLA+, some PETG-CF/PA-CF). Exact roller count/geometry not stated - inspect the Onshape CAD.
INDEXER / HOPPER Hopper/indexer feeding the drum shooter; the front of the hopper is attached to the intake so they fold down together, and it was cut down to fit in frame and respect extension limits. Detailed path geometry not stated; see the CAD.	**DRIVETRAIN** Carried over from Flint. Drivetrain type is not stated in the thread - unconfirmed (do not assume); see the Onshape CAD.
ENDGAME A deployable polycarbonate blocker (defensive) doubles as the folded-down front of the hopper when stowed; it was redone between Worlds and MROC after shearing the poly in Curie Finals 2, widened, and held down by surgical tubing looped to the bash bar when deployed. Climber/traversal endgame not described - unconfirmed.	**AUTONOMOUS** Won an Autonomous Award at the Mukwonago district event; specific routines are not enumerated in the thread.
VISION Not mentioned in the release thread - unconfirmed; check the CAD/config.	**PUBLIC CAD / DESIGN** 2026 CAD (Onshape, MROC config)

LESSONS FOR STUDENTS

- A textbook REBUILT case study: a district team rebuilt nearly the entire robot mid-season (fixed front shooter 'Flint' -> 17 in rear drum shooter 'Steel'), keeping only the drivetrain and intake pivot, then won the Curie Division at Worlds.
- Motor selection driven by weight: 4x Kraken X44 on the shooter instead of X60 for power density - similar peak power at a 40 A limit but ~2 lb lighter across four motors, decisive on a weight-constrained rebuild.
- Current budgeting up front: ~200 A total per shot, split ~120 A shooter / ~80 A rollers+indexer, sized to avoid brownouts.
- Integrating defense into structure: a deployable poly blocker that also serves as the folded-down front of the hopper; iterated (widened, surgical-tubing retention) after it sheared under match loads.
- Practical 3D-printed rollers/pulleys: eSun PLA+ (some PETG-CF/PA-CF), solid walls, large layer height, hex-bore inserts, printed upright, with exposed parts protected and swapped frequently.

Onshape CAD: https://cad.onshape.com/documents/db03fec505ded9ba473e7325 · Post-season thread + CAD (CD) · CAD (Onshape, MROC) · Statbotics 1714 (2026) · The Blue Alliance 1714 (2026)

#259 (study pick) **5847 Ironclad** (2026 comp bot) — Bradley, IL — **EPA 149** — 31-13

First-year Open Alliance team that won the Bluff Country Regional with turret + spindexer + adjustable hood

SHOOTER Hooded shooter with an adjustable hood; the hood was a key iteration point - broke two before settling on a wide-bottomed, more rigid design	**INTAKE** Roller intake (prototyped alongside shooter/spindexer/hopper using KrayonCAD concepting)
INDEXER / HOPPER Spindexer feeding the shooter, plus a hopper; tuned in code for spindexer rotation/feeding	**DRIVETRAIN** Swerve, Kraken X60 drive + Kraken X44 steer, top-mount 'brainpan' electronics; built to shoot-on-the-move
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed (first-year Open Alliance Regional winner).
VISION Limelight for turret aiming + pose; code adds voltage/supply/torque inputs and tunes intake, feed, shoot, spindexer, hood angle, and turret position	**PUBLIC CAD / DESIGN** Onshape (2026 robot)

LESSONS FOR STUDENTS

- A hood failure-and-redesign cycle (broke two, then widened the base) - a concrete structural-iteration lesson.
- Design prototypes for adjustability/versatility from one set of core parts (sliding-hood compression plate).
- Limelight-based turret aiming with code-tuned mechanism setpoints - an accessible vision approach for a mid-tier team.
- A first-year Open Alliance team documenting its way to a regional win.

Onshape CAD: https://cad.onshape.com/documents/b3b80a0c9175b5a0ee25425b · CD build thread · Onshape CAD · Behind the Bumpers

#269 **3928 Team Neutrino** Halcyon — Ames, IA — **EPA 147** — 26-16

Tall spindexer-turret architecture with a full-width linear (rack-and-pinion) intake; Iowa Regional winner

SHOOTER Flywheel shooter on a 360+ degree turret atop a tall spindexer tower	**INTAKE** Full-width LINEAR intake driven by a 10-DP rack and pinion (tooth profile modified to cut with a 1/8in end mill, inspired by 3467/6328); racks went polycarb -> 2x .2in MAXComposite + 1x .1in aluminum for tooth strength
INDEXER / HOPPER Tall spindexer feeding the turret; ~80-fuel hopper	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed (Iowa Regional winner).
VISION Unconfirmed (turret-aim code released)	**PUBLIC CAD / DESIGN** Full CAD release (Onshape) - first comp robot fully in Onshape

LESSONS FOR STUDENTS

- A rack-and-pinion linear intake gives a controlled straight-line deploy; 10-DP teeth modified for a 1/8in end mill make them CNC-able.
- Material selection is iterative: polycarb rack teeth sheared, so they moved to a MAXComposite + aluminum laminate - test under real load.
- Switching to Onshape pays off with parametric top-down design + FeatureScript despite transition hiccups.
- Plan for wear: turret wire sleeving wore quickly and became a per-event replacement.

CAD release: https://www.chiefdelphi.com/t/team-3928-team-neutrino-2026-cad-release/521183 · CD CAD release · Statbotics

#273 (study pick)

4561 TerrorBytes Cupid

Research Triangle Park, North Carolina, USA

EPA 146

46-17

No-turret double shooter fed by twin spindexers; the drivetrain itself aims, using odometry and Limelight MegaTag2.

SHOOTER

Double shooter, no turret. The team states they did not settle on a turret design and instead align the whole robot to the hub. The aiming method is stated in the thread: compute the field-relative angle to the hub from the x and y displacement, then subtract the robot's current heading to get a robot-relative target angle, with odometry zero facing the opposite alliance wall. Limelight MegaTag2 plus the gyro feed a WPILib SwerveDrivePoseEstimator through addVisionMeasurement. Flywheel specifics, hood geometry, motors and ratios are not stated in the build thread; read them from the CAD and code.

INTAKE

Prototyping settled on a single 3 in polycarbonate roller with WCP silicone tubing, which was enough to bring fuel in without a kicker roller or kicker plate provided compression stayed high; at lower roller speeds the fuel would not clear the bumper. The intake pivots, and constant-force springs retract the hopper when it pivots back rather than the intake pulling it. The pivot gearbox bound on first assembly from tight tolerances and was fixed by drilling out the motor holes and greasing it.

INDEXER / HOPPER

Twin spindexers, a layout the team calls a 'twindexer' after 7407's term, feeding the double shooter. Prototyping compared a spindexer against a roller floor: the spindexer needed a disruptor roller to stop fuel 'iglooing' above the main wheel and needed consistent compression, while the roller floor worked best on a slight decline with zip ties spaced along the rollers so the balls did not spin in place. An early twindexer prototype peaked at 15 fuel per second and averaged about 10.3. The hopper is polycarbonate, reinforced with aluminium L channel after the first plate proved floppy, with a supporting 1x1 removed and nut strips replaced by zip ties to save weight.

DRIVETRAIN

SDS MK5n modules, the team's first season on them, in a 30 x 25 in footprint chosen to fit the double spindexer while keeping hopper volume and a wide intake. The frame uses MAXTubes to carry the double shooter with clean wire routing, and a lightened aluminium bellypan stiffens it for a high-contact game. All electronics are mounted inverted beneath the drivebase, as on their 2025 robot. Bumpers are four-piece, which the team says puts each impact straight into the frame rather than into a gap or a 90 degree joint.

ENDGAME

Climber built and working. It was tested early without a drivetrain at 26 lb of load, and by the first event they were running a preload climb auto. The team notes that climbing during autonomous did not contribute to winning the auto period, but took the points anyway.

AUTONOMOUS

Started the first event without an auto and developed a preload climb auto on site. By Elon they were running a midfield auto consistently, which the team attributes to programming time rather than hardware. Won the Autonomous Award at Wake County #1.

VISION

Limelight with MegaTag2, fused with gyro rotation into a WPILib SwerveDrivePoseEstimator. New Limelight mounts before the Elon event removed the alignment problems seen at Wake County; the team names that as the main thing that had been holding back match performance.

PUBLIC CAD / DESIGN

Onshape CAD (main assembly) via the Open Alliance build thread

LESSONS FOR STUDENTS

- The clearest worked example in this set of aiming without a turret. The thread contains the actual math for pointing the drivetrain at the hub from odometry, and a teammate reply walks another team through it.
- Prototyping conclusions are written down as rules rather than outcomes: a spindexer needs a disruptor roller and consistent compression; a roller floor wants a slight decline and zip ties to stop balls spinning in place.
- High intake compression removed the need for a kicker roller or kicker plate. One well-chosen parameter deleted a subsystem.
- Driving over the bump works best diagonally, with bumpers higher than 1 in off the ground, and added drivetrain weight reduces airtime. That is a testable finding, not an opinion.
- A full-season Open Alliance thread including the failures: battery drainage and brownouts early, a binding pivot gearbox, a floppy hopper plate, and what each fix was.
- Season result: Autonomous Award and finalist at Wake County #1 at 14-4, Impact Award and finalist at Elon, then Impact Award, the FIRST Leadership Award and a Finals win as Alliance 2's first pick at NCCMP.

CD thread: https://www.chiefdelphi.com/t/4561-the-terrorbytes-2026-build-thread-open-alliance/508623 · 2026 Build Thread (Open Alliance) · Statbotics 4561 (2026) · The Blue Alliance 4561 (2026)

#282 2582 PantherBots Cobra

Lufkin, TX

EPA 144

37-20

Adjustable-hood dual-motor launcher + hopper feeder; high-volume scorer; qualified Worlds

SHOOTER Adjustable-hood launcher, two flywheel motors, linear-actuator hood read by an absolute encoder; high throughput (~260 fuel/match claimed)	**INTAKE** Floor intake feeding the hopper (works with hopper to empty fuel quickly)
INDEXER / HOPPER Hopper + feeder delivering fuel to the dual-motor launcher	**DRIVETRAIN** Swerve; drivetrain assembled early; vision still in progress late-Feb
ENDGAME Side L1 climb - the CG offset from the hook needs ~0.6 friction or a dedicated gripper.	**AUTONOMOUS** Adjustable-hood dual-motor launcher with an absolute-encoder hood for repeatable shot angles; auto routines not detailed.
VISION Unconfirmed (integration in progress during build)	**PUBLIC CAD / DESIGN** Onshape CAD + GitHub (Open Alliance blog)

LESSONS FOR STUDENTS

- A linear-actuator hood with an absolute encoder gives repeatable, tunable shot angles for variable distance.
- Side L1 climbing creates a tricky moment - CG offset from the hook needs ~0.6 friction or a dedicated gripper.
- Dual launcher motors must be monitored - a single over-current motor silently halves your shooting (watch current draw).
- Document openly (GitHub/Onshape/blog) to keep iteration disciplined through a Worlds-qualifying season.

CD thread: https://www.chiefdelphi.com/t/2582-pantherbots-build-blog-open-alliance-2026/509632 · CD build blog · Reveal 'Cobra'

#323 1778 Chill Out HAILSTORM

Washington, USA

EPA 134

25-19

'HAILSTORM': compact turret shooter fed by a spindexer; motor + single absolute-encoder turret; Kotlin code; swerve

SHOOTER Turret-mounted launcher (described as a compact turret assembly). Per the thread, the team weighed a 'dye rotor' but rejected it as too complex and chose the turret/spindexer path for simplicity with a high ceiling. Flywheel motor/ratio specifics not stated in the thread - read from the Onshape model.	**INTAKE** Ground intake; the team noted a mid-season intake redesign that delayed their practice-field time. Specifics not detailed in the thread - see CAD.
INDEXER / HOPPER Spindexer feeding the turret. The team chose a spindexer over a dye rotor as the simplest solution with a high ceiling. (The cone was finished off-CAD with cat-tongue grip-tape strips, heat-shrunk to conform for extra grip.)	**DRIVETRAIN** Swerve (module/gearing not stated in the thread - see CAD).
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Turret zeroing via a motor (relative) encoder + one absolute encoder; auto routines not detailed.
VISION Not named in the thread - the turret aims via encoders rather than a stated camera; treat as Unconfirmed and check the CAD/code repo.	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape) + code (GitHub, Kotlin)

LESSONS FOR STUDENTS

- Pragmatic turret zeroing: the motor (relative) encoder handles continuous tracking, and a single absolute encoder geared ~5:1 supplies a re-zero value while shooting; closest-region logic plus manual re-zero buttons cover encoder drift - a cheap alternative to a full CRT/absolute setup they couldn't tune in time.
- Archetype selection as risk management: they rejected a dye rotor for a spindexer specifically because it was the simplest path with a high performance ceiling.
- A public Kotlin codebase is a useful non-Java reference for teams exploring JVM languages.
- A recovery case: early power-consumption issues, then a hardening/tuning focus; the team rates the result among its better robots by DCMP.

Onshape CAD: https://cad.onshape.com/documents/c527fb2ecdd377cbb954f474/w/b89f138221c8d9223c28e242/e/c4daa58f8630b2239d8ee484 · CAD & Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

#359 48 Team E.L.I.T.E. xtremachen 29

Ohio

EPA 128

29-23-1

Turret spindexer shooter, fully in-house manufactured, fast-iterated vision/SOTM; Midwest Regional winner

SHOOTER Turret-mounted flywheel with shoot-on-the-move	**INTAKE** Roller intake with a quick-change roller (a bent roller is a fast pit swap); stiffened + de-jammed after the first regional
INDEXER / HOPPER Spindexer (cone geometry inspired by 4145)	**DRIVETRAIN** Swerve (module unconfirmed)
ENDGAME Climber present but iterated for reliability (they considered removing it between events).	**AUTONOMOUS** Two Limelight 4s + pose + shoot-on-the-move brought online in ~8 days between events.
VISION 2x Limelight 4 for AprilTag pose + shoot-on-the-move; stood up in ~8 days mid-season	**PUBLIC CAD / DESIGN** Full CAD release (Onshape) - first Onshape season, 100% in-house parts

LESSONS FOR STUDENTS

- Vision is a software sprint you can win late: two Limelight 4s + pose + SOTM working in ~8 days between events.
- Design for fast field repair - a quick-change intake roller turns a bend into a pit swap, not a dead robot.
- Iterate event-to-event with a clear list (stiffen intake, fix jamming, even remove the climber for reliability).
- Borrowing a subsystem detail (spindexer cone from 4145) beats reinventing it.

CAD release: https://www.chiefdelphi.com/t/frc-48-team-e-l-i-t-e-xtremachen-29-2026-cad-release/521258 · CD CAD release · Statbotics

#377 6672 Fusion Corps Bastion

Texas, USA

EPA 124

19-26

'Bastion': turreted flywheel shooter with kicker feed, floor intake and indexer, on CTRE swerve; PhotonVision AprilTag targeting. Full Onshape CAD + AdvantageKit-style code released.

SHOOTER Turreted flywheel launcher. The released code exposes separate Shooter, Turret and Kicker subsystems, each driven through CTRE TalonFX controllers (Kraken/Falcon-class motors; specific motor models unconfirmed - see the CAD BOM). The dedicated Turret subsystem means the launcher rotates independently of the chassis (shoot-while-aiming); a Kicker meters game pieces into the flywheel. Flywheel wheel/hood geometry and gear ratios are not stated in the release post - read them from the Onshape model.	**INTAKE** Floor intake driven by a CTRE TalonFX controller (Intake subsystem in code). Roller/geometry detail not described in the release post - inspect the Onshape CAD.
INDEXER / HOPPER Indexer driven by a REV SPARK controller (NEO-class motor; model unconfirmed). Sits between the intake and the kicker/turret feed. Path geometry not described in text - see the CAD.	**DRIVETRAIN** Swerve. Code uses CTRE Phoenix 6 (generated TunerConstants, TalonFX/TalonFXS module controllers, PhoenixOdometryThread) with a selectable NavX or Pigeon2 gyro, and PathPlanner autonomous. Module make/ratios unconfirmed in the post - see the CAD.
ENDGAME No climber/endgame subsystem appears in the released code; endgame mechanism (if any) unconfirmed - check the Onshape CAD.	**AUTONOMOUS** PathPlanner-based autonomous routines (pathplanner deploy folder in the repo); specific paths/sweeps not enumerated in the release post.
VISION PhotonVision for AprilTag localization (VisionIOPhotonVision + photonlib vendordep in the released code). Camera count/placement not stated in the post - see the CAD/config.	**PUBLIC CAD / DESIGN** 2026 CAD Release (Onshape)

LESSONS FOR STUDENTS

- An AdvantageKit-style codebase: every mechanism (drive, shooter, turret, kicker, intake, indexer, vision) is split into a subsystem + hardware IO interface + sim implementation. A template for teams learning that architecture.
- A separate Turret subsystem decoupled from the drivetrain shows how to structure code for a shoot-on-the-move turreted launcher.
- PhotonVision + Phoenix 6 swerve + PathPlanner is a widely reusable, fully-public localization/auto stack for a mid-resource team.
- Mixed motor-controller ecosystem in one robot (CTRE TalonFX on most mechanisms, REV SPARK on the indexer) - a realistic example of using the right controller per mechanism.

Onshape CAD: https://cad.onshape.com/documents/c8f7278d84c0d231b4ae782c/w/32ba4de6819aba4f14857523/e/4f4804f7c4c66c747eefa67f · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics 6672 (2026) · The Blue Alliance 6672 (2026)

#399 **1676 Pascack PI-oneers** Cobra Pi
Montvale, NJ

EPA 123
35-24-1

High-capacity hopper shooter known for big event-to-event upgrades; ~80-ball usable storage

SHOOTER Flywheel shooter (turret detail unconfirmed); continuous upgrades each event	**INTAKE** Roller intake (motors unconfirmed); throughput slows once the hopper is heavily loaded
INDEXER / HOPPER Large hopper: ~80 balls before fuel weight bogs the intake, ~90-95 at overflow	**DRIVETRAIN** YAGSL swerve, Kraken X60 drive + Kraken X44 steer, CANcoder, navX
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed; large event-to-event upgrades.
VISION 3x Limelight (left/right/front), MegaTag pose estimation	**PUBLIC CAD / DESIGN** CAD + code release (Onshape)

LESSONS FOR STUDENTS

- Add-on Delrin 'skis' under the drivebase help the robot ride over the field bump and shield swerve motors - a cheap fix for a known hazard.
- Know your REAL capacity limit (~80 usable) vs physical (90-95) - design to the usable number.
- Aggressive event-to-event upgrades drove a steep improvement curve.

CAD release: https://www.chiefdelphi.com/t/1676-cad-release-2026/520302 · CD CAD + code release · Team mechanical page · Released code (practice chassis)

#399 **3175 Knight Vision** Sisyphus
Michigan, USA (FiM)

EPA 123
33-24

Four-bar intake, roller hopper floor with 3D-printed structural plates, shooter-and-tower assembly; drivetrain unconfirmed.

SHOOTER
The released assembly holds one subassembly named 'Shooter + Tower'. A commenter refers to the shooter side plates carrying cantilevered aluminium Limelight mounts. Wheel sizes, motors, ratios and hood travel are not stated in the thread; read them from the CAD.

INTAKE
The released assembly names the intake '4 Bar Intake'. The team states the intake was rebuilt after their first event, at the same time as the hopper plate change. Some hopper roller types exist to clear the coaxial intake. Motors, compression and roller sizes are not stated in the thread; read them from the CAD.

INDEXER / HOPPER
Hopper floor built on The Thrifty Bot dead-axle roller system, which the team already had in stock. The stock stub roller could not be removed without taking its whole plate off the robot, because the shaft is unthreaded into a nut. The team kept the TTB hubs and replaced the shaft with a 1/2 in hex shaft, using a shoulder screw for the front bearing, so rollers swap out in place; one side is tapped 10-32 and the other 1/4-20. That modification is not shown in the released CAD. Stock TTB rollers accept only 24-tooth pulleys, so WCP SplineXL hubs were used to run larger 3D-printed pulleys and vary the speed of alternate rollers on the hopper floor. Several roller types are used so balls do not treadmill. The hopper plates are 3D-printed structural plates, adopted after the first event for weight and rigidity; the team recommends an isogrid pocketing pattern and reports it was the most rigid pattern in their experience.

DRIVETRAIN
Unconfirmed. The released assembly holds a 'Drivetrain' subassembly; the thread does not state the drivetrain type. Read it from the CAD.

ENDGAME
Not stated in the thread. No climber or endgame mechanism is named in the released assembly tree.

AUTONOMOUS

VISION
Unconfirmed. A commenter describes aluminium Limelight mounts cantilevered off the shooter plates and asks about vibration; the team does not answer and does not name a vision system. Camera count and model are unconfirmed.

PUBLIC CAD / DESIGN
2026 CAD (Onshape)

LESSONS FOR STUDENTS

- A COTS dead-axle roller can be made field-serviceable by changing the shaft, not the whole system: a 1/2 in hex shaft plus a shoulder screw lets one roller come out without removing its plate.
- Check what a COTS roller constrains before committing to it. The TTB stub roller accepts only 24-tooth pulleys, which blocked per-roller speed changes until SplineXL hubs and printed pulleys were added.
- Varying the speed of alternate hopper rollers, and mixing roller types, is one way to stop balls treadmilling in a hopper floor.
- 3D-printed structural plates replaced the original hopper plates mid-season for weight and rigidity; isogrid pocketing was the team's most rigid pattern.
- The thread carries no code, binder, motor list or ratios. Treat this release as a hardware reference to read in CAD, not a documented design.

Onshape CAD: https://cad.onshape.com/documents/8fa47d58e24d11fb99ea01f6/w/b35f6644b105cf745abd0a5c/e/4e87fd7cd5b50d294798ca46 · CAD release thread (CD) · 2026 CAD (Onshape) · Custom stub roller doc (Onshape) · Statbotics 3175 (2026) · The Blue Alliance 3175 (2026)

#381 (study pick) **2079 4-H ALARM Robotics** Deadshot — **EPA 122**
Franklin, MA — 26-20

Drum shooter (no turret) with a compression-tuned over-bumper intake + roller indexer

SHOOTER Drum shooter (the team states it was satisfied with it); planned to grind the hood for a higher-arching shot; SOTM not yet implemented	**INTAKE** Over-bumper intake whose compression was too high - couldn't copy another team's geometry, had to run their own compression tests; broke several times and was reinforced with aluminum plates
INDEXER / HOPPER Indexer rollers feeding the drum; planned smaller indexing rollers to raise throughput	**DRIVETRAIN** Kraken X60 swerve (TalonFXS modules)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Drum shooter; shoot-on-the-move not yet implemented (planned).
VISION 4x PhotonVision cameras (back-left, back-right, side-left, side-right) - PhotonVision, not Limelight	**PUBLIC CAD / DESIGN** Onshape CAD + GitHub (Open Alliance)

LESSONS FOR STUDENTS

- Intake compression must be tuned to YOUR robot - copying another team's spacing doesn't transfer; run your own ball-compression tests.
- Design for durability: an intake that breaks repeatedly needs structural reinforcement, not just tuning.
- Smaller indexing rollers can raise throughput - small geometry changes meaningfully affect ball flow.
- Borrow hopper-capacity ideas from top teams (passive net like 5687, vertical active expansion like 1678).

CD thread: https://www.chiefdelphi.com/t/frc-2079-alarm-robotics-2026-rebuilt-build-thread/509762 · CD build thread · GitHub · Robot code (DeadShot)

#496 (study pick) **4392 The Deceivers** (2026 comp bot) — **EPA 107**
Brimley, MI — 24-20

Small (17-student) team that built an ambitious turret shooter with a star-wheel intake

SHOOTER Turret: dual Kraken X60 on a 1:1 shooter axle, two SDS brass flywheels + a single 4in Fairlane 60A wheel; adjustable hood with a 1in hood roller geared 2:1	**INTAKE** Over-bumper 4in compression wheels + star wheels (no kicker bar needed); deployed by dual mirrored rack-and-pinion (racks = 4 epoxied laminations of .120 polycarb, PETG pinions)
INDEXER / HOPPER Roller hopper with a passive funnel + a kicker that flicks balls into the shooter (254-2017 style); chose roller hopper over spindexer after prototyping both	**DRIVETRAIN** Swerve, Kraken X60 drive, 4.2:1, 38mm wheels, ~5.5 m/s, 0.55m track width
ENDGAME Climber not detailed (small 17-student team).	**AUTONOMOUS** Auto routines not detailed.
VISION 4x Limelight (one..four), MegaTag2	**PUBLIC CAD / DESIGN** Onshape (2026 document)

LESSONS FOR STUDENTS

- A star-wheel + compression-wheel over-bumper intake can work WITHOUT a kicker bar.
- Rack-and-pinion intake deploy with laminated-polycarb racks - fabrication on a budget.
- Roller-hopper vs spindexer: they prototyped both and documented why they picked rollers.
- Honest self-critique (3 lb of brass flywheel) models good engineering process for students.

Onshape CAD: https://cad.onshape.com/documents/589237fcf2364719e491ed4b · CD Open Alliance thread · Onshape CAD · Website · Robot code

#528 **1577 Steampunk** Cerberus
Israel
EPA 107
21-20

Wide-intake high-capacity turret shooter, fully student-built; reached Newton Division

SHOOTER Flywheel turret shooter, ~19 BPS sustained	**INTAKE** 5-balls-wide intake
INDEXER / HOPPER Large hopper, ~75-ball capacity	**DRIVETRAIN** Swerve at an 'R2' (MAXSwerve-class) ratio; max frame perimeter, max-weight robot
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Auto routines not detailed; ~19 BPS sustained throughput.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** CAD + Technical Binder (Onshape)

LESSONS FOR STUDENTS

- A wide intake (5 across) + a big ~75-ball hopper makes a high-throughput cycle machine - width and capacity are force multipliers.
- Fully in-house, student-led manufacturing is achievable; pair it with a written technical binder so the reasoning survives to next year.
- Pick a known-good swerve ratio ('R2') instead of over-customizing, and spend the saved effort on scoring mechanisms.

CAD release: https://www.chiefdelphi.com/t/steampunk-1577-2026-cad-and-technical-binder-release/521240 · CD CAD + tech binder release · Statbotics

#710 **5811 The BONDS** Lektor
Ohio
EPA 85
15-16

720-deg turret shooter with a 3D-printed spindexer and a 1678/27-style slapdown intake; mid-season rebuild to add a hopper

SHOOTER Flywheel on a 720-degree turret rotating on a 7in X-Contact bearing; praised for tight turret wire management + ball-exclusion shielding	**INTAKE** '27/1678-inspired' slapdown intake; later swapped to a rebuilt 'Buckeye' 4-bar intake mid-season
INDEXER / HOPPER 3D-printed spindexer; rebuilt between Wk3 and Wk5 to move the intake to the short side, freeing room to add a hopper	**DRIVETRAIN** Swerve, 32x23.5in base
ENDGAME Climber not detailed; a mid-season rebuild added a hopper.	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed	**PUBLIC CAD / DESIGN** CAD release + reveal video (Onshape)

LESSONS FOR STUDENTS

- A mid-season rebuild is workable: moving the intake from the long side to the short side opened room for a hopper and, per the team, 'significantly improved' offense.
- Borrow proven geometry - their slapdown intake is openly modeled on 1678/27, a fast path to a working mechanism.
- A 720-degree turret on a big X-contact bearing needs disciplined wiring + a ball-exclusion shield.
- 3D-printed spindexers are a low-cost, iterable storage option when you can't machine custom parts.

CAD release: https://www.chiefdelphi.com/t/team-5811-the-bonds-2026-robot-cad-release/520901 · CD CAD release · Statbotics

#730 **2183 Tigerbots**
Louisiana, USA

EPA 83
18-18

720-deg passive central front turret; beam-break-controlled dual-roller polybelt serializer; large hopper; MK4n swerve sized to fit under the trench

SHOOTER Shooter rides a central front turret with 720 deg of travel achieved via a passive mechanism (no re-zeroing); wheel/motor specifics not stated in the thread - see CAD.	**INTAKE** Ground intake feeding the large hopper (specifics not detailed in the build thread - see CAD).
INDEXER / HOPPER Center serializer with two polybelt roller stacks that are INDEPENDENTLY controlled (not belted together) using beam-break sensors: rather than simply alternating, each roller is driven based on whether a ball is detected - sometimes both roll inward, changing behavior when a beam break trips - to de-interlock FUEL and funnel the wide hopper into one stream. (Corrected 2026-07-06 per team CAD/Code lead: earlier text wrongly assumed the two rollers were belted together and merely alternated.) Reached via 4+ prototypes - a spindexer (too sensitive to gravity/geometry) and a two-wheel track (balls interlocked) were rejected; V1 (top belt only) ran the Oklahoma and Magnolia regionals before the final version.	**DRIVETRAIN** SDS MK4n swerve; overall height constrained to drive under the trench, which motivated the low central turret architecture.
ENDGAME Climber not detailed in the public release.	**AUTONOMOUS** Auto routines not detailed.
VISION Unconfirmed (not stated in the build thread).	**PUBLIC CAD / DESIGN** Full CAD (Onshape) + subsystem build thread

LESSONS FOR STUDENTS

- Independently-driven serializer rollers with beam-break sensing (rather than one belted-together set) let the robot vary each roller's behavior by ball state to clear and prevent jams.
- Their thread shows the full failure path (spindexer -> track -> belted serializer): documenting WHY prototypes failed is as instructive as the final design.
- A passive 720-deg turret gets shoot-anywhere coverage without slip rings or complex wiring - clever low-budget engineering.
- Packaging constraints (fit under the trench with MK4n modules) drove the whole architecture - constraints first, mechanism second.

Onshape CAD: https://cad.onshape.com/documents/0f18097cc68763333bbf9187/w/ec4ab9bf696dddb823f66e6c/e/e6c97ba1f43ead9fa84d0197 · CD build thread (turret + serializer) · Full CAD (Onshape) · Statbotics

#844 (study pick)	**1601 Quantum Samurai** Linguist New York, USA	**EPA 75** 15-11

'Linguist': 360-degree turreted flywheel shooter with adjustable hood, deployable over-the-bumper rack-extension intake, tower-fed hopper/indexer; WCP X2C swerve

SHOOTER Turret-mounted. 2x Kraken X60 belted to a 4 in OD Colson wheel via shaft at 11:18; two 1.5 lb brass flywheels belted to the same Colson shaft at 18:18 to smooth and hold rotational momentum. FUEL clearance (main-plate diameter) is 6.51 in; each flywheel retained by the side plates plus 1/8 in aluminum plates. Adjustable ABS hood driven by a Kraken X44 through an ABS double-herringbone rack and pinion at >200:1 (step-up gears), giving 21-50 deg of hood travel; anti-flex pins extend into the rack, ABS e-chain manages wiring.	**INTAKE** Deployable over-the-bumper intake on the front two X2C modules. Structure: three 0.125 in polycarbonate panels joined at 90 deg with aluminum top brackets and a 1x1 0.125 in aluminum crash bar for impact. Rollers: a 2 in polycarbonate roller in silicone tubing on a 3D-printed 24T hub (WCP stub shaft) plus a 1.5 in polycarbonate roller in CatTongue grip tape on an 18T hub, belted together and to a kicker roller via a Kraken X60 at 12:18. A kicker improves FUEL compression consistency.
INDEXER / HOPPER Tower-fed hopper/indexer. Indexer = two 0.25 in polycarbonate side panels 22.5 in apart with 0.125 in polycarbonate vector backing panels; two vectored 2 in intake-roller shafts plus center compliant wheels. Before reaching the tower, FUEL is agitated by 4x 1.25 in dia aluminum hotdog rollers (20.5 in long) on 20T WCP stub rollers; all shafts belted 12:24 to a Kraken X60. The tower funnels FUEL up to the turret and is reinforced with 1/4 in polycarbonate supporting plates.	**DRIVETRAIN** 27x27 in frame on four WCP X2C swerve modules. Double-bellypan design: two aluminum bellypans mount electronics (one for the PDP, one for the RoboRIO / Pigeon IMU / CANivore), with a bottom Slippery UHMW skid panel spanning the base for debris protection and as an anti-beaching surface. 3D-printed TPU battery sock retains the battery.
ENDGAME Endgame/climber not described in the released tech binder; check the Onshape CAD.	**AUTONOMOUS** Autonomous routines not detailed in the binder; software detail lives in the team's resources.
VISION Three Limelights for full-field localization: two mounted atop the X2C modules nearest the turret tower and a third against a hopper support stock - the robot always knows its field pose for turret aiming.	**PUBLIC CAD / DESIGN** 2026 CAD + Tech Binder (Chief Delphi)

LESSONS FOR STUDENTS

- The turret is built around an X-contact bearing with a 3D-printed 184T ring gear driven by a 24T aluminum pinion (38.334:1) off a Kraken X60 / MAXPlanetary 5:1 - a fully student-documented turret others can copy.
- A two-stage flywheel (driven 4 in Colson at 11:18 plus 1.5 lb brass inertia flywheels at 18:18) is a clean way to smooth shot-to-shot energy without a heavier single rotor.
- End-to-end documentation for a mid-tier team: every one of the six mechanisms (turret, shooter, chassis, tower, hopper/indexer, intake) has a materials-and-ratios writeup plus CAD. Study material for newer teams.
- Process honesty: the binder credits CAD overviews from 1323 and 1678 just before their first regional, showing how design reviews from veteran teams shaped quick fixes.

CD thread: https://www.chiefdelphi.com/t/522131 · CAD + Tech Binder release (CD) · Statbotics · The Blue Alliance

#932 (study pick)

6381 Red Raider Robotics (2026 comp bot)

Sheboygan, WI

EPA 70

24-23

Simplicity-first single shooter (NO turret), modular per-subsystem Onshape docs

SHOOTER Single fixed shooter (no second shooter, no turret) - chose one for accuracy, judged they can move balls fast enough; turret deferred to the offseason due to inexperience	**INTAKE** Over-bumper / floor intake (its own Onshape doc; outtake variants prototyped on YouTube)
INDEXER / HOPPER Dedicated indexer subsystem in its own Onshape doc feeding the single shooter	**DRIVETRAIN** Chassis subsystem in its own Onshape doc (swerve expected)
ENDGAME Climber not detailed (simplicity-first, no turret).	**AUTONOMOUS** Auto routines not detailed; one accurate fixed shooter by choice.
VISION Unconfirmed (no clear vision pipeline documented)	**PUBLIC CAD / DESIGN** Onshape - one doc per subsystem (Shooter/Indexer/Intake/Chassis/Climber)

LESSONS FOR STUDENTS

- Scope to your team's experience: deliberately skipping a turret keeps a developing team reliable rather than over-reaching.
- One accurate shooter can beat two if you can feed balls fast enough - fewer mechanisms, fewer failure points.
- Splitting CAD into one Onshape doc per subsystem lets students work in parallel without conflicts.
- Defer ambitious mechanisms (turret) to the offseason as a learning project rather than risking the season.

Onshape CAD: https://cad.onshape.com/documents/e139ac1c610559bc8767c8da · CD build thread · Onshape subsystem docs

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#1109 (study pick)

5982 Merrill's Circuits of Destruction Doubloon

Michigan, USA

EPA 60

21-21

'Doubloon': WCP Competitive Concept-based scorer reworked for serviceability; PhotonVision

SHOOTER Based on the WCP Competitive Concept shooter (specifics not detailed in the release).	**INTAKE** Quick-change intake: every roller runs on standard hex shafts so a broken roller swaps in under a minute; geometry revised for more consistent pickup.
INDEXER / HOPPER Lightweight net hopper replacing the CC's heavy extendable hopper -- lighter and more durable.	**DRIVETRAIN** Swerve, built from the WCP Competitive Concept base (module specifics unconfirmed).
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** PhotonVision; auto routines not detailed.
VISION PhotonVision -- integrated for on-field targeting (not modeled in the CAD).	**PUBLIC CAD / DESIGN** Full CAD (Fusion 360)

LESSONS FOR STUDENTS

- Adapting a COTS concept (WCP Competitive Concept) and optimizing it for serviceability -- quick-change hex-shaft rollers -- is a legitimate, time-efficient design path.
- Replacing a complex extendable hopper with a simple net cut weight and improved durability: simplicity as a design win.

Team site: https://a360.co/3NHiaZ3 · CD CAD release · Statbotics

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#1358 (study pick)

6333 Scavenger Robotics Marion

Massachusetts, USA (NE)

EPA 49

11-17

Small-team turret shooter with a spindexer, built by 7 students on about a 20k total budget; the team's first turret and first spindexer.

SHOOTER The team's first turret robot. They report reaching a consistent 3 fuel per second by the end of the season and name the spindexer size, not the shooter, as the throughput limit. Flywheel or drum type, hood geometry, motors and ratios are not stated in the release post; read them from the CAD and the technical binder.	**INTAKE** Not described in the release post. The detail is in the released CAD and technical binder; unconfirmed here.
INDEXER / HOPPER Spindexer, the team's first. They state its small size bottlenecked throughput, which is the constraint that held their fuel rate at about 3 per second.	**DRIVETRAIN** Differential (tank) drive, not swerve. This is recorded as a correction in the project's own notes after an earlier entry mislabelled it; the release post itself does not state the drivetrain, so confirm it in the CAD. Marion is the only non-swerve robot among the study picks.
ENDGAME Not described in the release post; unconfirmed.	**AUTONOMOUS** Not described in the release post; unconfirmed. The team's own summary is that most things were manually controlled.
VISION Minimal. The team states the robot did not have much going for it in software and that most functions were manually controlled, and names the lack of good vision-based automation as one of the two things that held the robot back.	**PUBLIC CAD / DESIGN** Onshape CAD + technical binder

LESSONS FOR STUDENTS

- A resource reference point that the rest of this list does not give you: 7 students, 4 mentors, and about 20k for the robot, event fees, shirts and pizza together.
- The team names its own limits plainly: mechanically robust and reliable, weak in software, throughput capped by a spindexer that was too small. That is more useful to copy than a highlight reel.
- First turret and first spindexer in the same season, reaching a consistent 3 fuel per second. A realistic target for a team attempting either mechanism for the first time.
- The only tank-drive robot among the picks. Swerve is common at the top of this list and is not a requirement for releasing studyable CAD and documentation.
- They released CAD and a binder for a robot that did not reach DCMP or Worlds, and won their second Imagery Award at the NE WPI event.

CAD release: https://www.chiefdelphi.com/t/team-6333-robot-reveal-and-cad-release/518178 · Robot reveal and CAD release (CD) · Statbotics 6333 (2026) · The Blue Alliance 6333 (2026)

#1838 135 Penn Robotics (Black Knights) unconfirmed

Indiana, USA

EPA 38

24-38

Swerve, twin-turret twin-flywheel shooter; 254-2017-style floor-roller indexer, wide floor intake, omnidirectional side/center hang.

SHOOTER Twin flywheel shooters on two turrets, chosen for redundancy ('one is none, two is one' - keep firing if one jams or stalls). Each turret pairs a flywheel with an adjustable hood (about 15-45 deg hood range, turret elevation about 45-90 deg); the flywheel is either direct-driven or on a 180-degree gearbox to keep the motor parallel with the flywheel shaft. A prototype used a 4in green compliant wheel (softer neoprene-tread wheels shredded the game pieces). Motors/ratios: see the Onshape CAD.	**INTAKE** Wide floor intake - the frame was maximized on the intake/turret side for the widest possible intake. Battery and RIO were packaged under the intake to free hopper volume. Roller count/motors: see the CAD.
INDEXER / HOPPER Indexer modeled on 254's 2017 design, with team 118 also referenced. Silicone-wrapped 1in-OD 'floor rollers' geared up to keep constant upward pull on the fuel, plus 'kickup shelf' rollers near the turrets to raise balls-per-second and act as a 'jackhammer' to clear clogs. The layout is designed to stack about 3 fuel vertically across most of the robot.	**DRIVETRAIN** Swerve (the team describes 'a turret/swerve' architecture); 30x25in frame. Module brand/ratio beyond swerve is unconfirmed; the CAD has a 'DriveTrain135-2026' tab.
ENDGAME Omnidirectional hang (CAD 'Telescoping Arm' + 'Elevator'). Extending arms reach up one at a time to hang off the side or center of the ladder tower, so the robot does not block partners' side-hang spots; compliant wheels grip the bar to roll into place and reduce denting and sway. The battery is mounted under the hang to bias weight toward it.	**AUTONOMOUS** Not detailed in the thread; see the 2026-Rebuilt code repo (includes turret azimuth math).
VISION Custom machine-learning vision: their 'Southmoon' repo contains a trained model (best.pt and a best.mlpackage). The specific pipeline/product is unconfirmed - read the code.	**PUBLIC CAD / DESIGN** 2026 CAD (Onshape)

LESSONS FOR STUDENTS

- Twin-turret redundancy: 135 built two independent turret shooters so the robot can keep scoring if one fails - a concrete durability-vs-complexity trade for a scoring subsystem.
- Indexer throughput details: a 254-2017-style floor-roller indexer with silicone-wrapped 1in rollers plus 'kickup shelf' rollers near the turrets to raise balls-per-second and clear clogs.
- Omnidirectional hang: extending arms that side- or center-hang on the ladder with compliant wheels gripping the bar, designed not to block alliance partners' hang spots.
- A documented Open Alliance thread from a mid-tier team (local rank 1838, 22-40): included for the downloadable CAD and design log, not for competition results. Parameterized CAD via Onshape Variable Studios. 2025 FIN Lafayette Impact Award winner.

Onshape CAD: https://cad.onshape.com/documents/20202353e5ff6b6c6fe21c64 · Open Alliance build thread (CD) · 2026 CAD (Onshape) · Code (GitHub, 2026-Rebuilt) · Statbotics 135 (2026) · The Blue Alliance 135 (2026)

#1896 **157 The AZTECHS** (2026 comp bot)
Marlborough, MA

EPA 36
11-18

Modest on-field results; deeply documented vision + turret engineering

SHOOTER Two-motor single-wheel flywheel using 4in ThriftyBot urethane wheels at ~1in compression; adjustable hood 20.0-57.5 deg	**INTAKE** Roller intake; added a second Kraken X60 which greatly sped up intaking and reduced jams
INDEXER / HOPPER Polycarb-roller hopper slanted toward the uptake (grip via hockey tape / textured TPU); 254-style turret wiring; turret sensed by an effective 1:1 encoder over 360 deg	**DRIVETRAIN** Thrifty Narrow Swerve via the CTRE Swerve Project Generator; uses YAMS (Yet Another Mechanism System)
ENDGAME Climber not detailed in the release.	**AUTONOMOUS** Multi-camera PhotonVision on dual Orange Pi 5 (deeply documented vision); auto routines not detailed.
VISION PhotonVision 2026 on two Orange Pi 5 coprocessors (load split across ~4 ThriftyCams); 3 cams on the shooter platform for localization + 1 on the turret for hub alignment; custom AprilTag mounting plates	**PUBLIC CAD / DESIGN** Open Alliance build thread (CAD/links in-thread)

LESSONS FOR STUDENTS

- Multi-camera PhotonVision on dual Orange Pi 5 - splitting cameras across coprocessors, MJPEG-vs-YUYV trade-offs, per-Pi hostnames (a thorough vision study).
- Separate localization cameras (chassis) from a targeting camera (turret) - a clean architecture pattern.
- 254-style turret wiring + a 1:1 absolute encoder for a 360-deg turret.
- Adding a 2nd intake motor to cure jams - a simple, high-impact reliability fix. (Study for vision/turret, not on-field results.)

CD thread: https://www.chiefdelphi.com/t/frc-157-the-aztechs-2026-build-thread-open-alliance/509425 · CD build thread · Statbotics

Sources: Chief Delphi (reveals & Open Alliance build threads), David's Design Server / frcdesign.org (robot-pics-2026 & mechanism-examples), team CAD/code releases (Onshape, GrabCAD, GitHub), FUN "Behind the Bumpers", and Statbotics EPA & records (2026). "Unconfirmed" means a detail was not publicly published, not that it is absent.